

《Automotive Configuration》

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Lecture Schedule: 2nd Semester Week 1~8

第五章 柴油机燃油供给系统

Chapter 5 Fuel Supply System for Diesel Engine



Overview

5.1 Characteristics and Properties for Diesel Fuel

柴油及其使用性能

5.2 Composition of Fuel Supply System for Diesel Engine

柴油机供给系统组成

5.3 Injectors

喷油器

5.4 Plunger Injection Pump

柱塞式喷油泵

5.5 Distributor Injection Pump

分配式喷油泵

5.6 Speed Governor

调速器

5.7 Electronic-controlled Injection System for Diesel Engine

柴油机电控燃油喷射系统

5.8 Auxiliary Device

电控燃油喷射系统辅助装置

5.1 Characteristics and Properties for Diesel Fuel (柴油及其使用性能)

1. Ignition Quality ——cetane number (CN) (自燃性的指标——十六烷值 $C_{16}H_{34}$)

CN $\uparrow$ →Ignition Quality $\uparrow$ →Ignition Delay (着火延迟时期) $\downarrow$ →Starting and combustion behavior(冷启动及燃烧特性) $\uparrow$ →Exhaust and noise emissions (排放及噪音) $\downarrow$

CN $\uparrow\uparrow$ → Molecular Weight(燃料分子量) →Evaporability(蒸发性) $\downarrow$
→Viscosity(粘度) $\uparrow$ →Smoke Formation $\uparrow$ &Fuel Economy(燃油经济性) $\downarrow$ 。

cetane number ≥ 45 (Presently lies between 51 and 56)

2. Evaporability—Distillation(蒸馏程): Temperature range where diesel evaporates to a specific percentage

- 50% boiling point(50%馏出温度)presents the average evaporability(平均蒸发性能):
50% point $\downarrow$ →quick evaporation →mixture formation $\uparrow$; however, mixture formation gets too fast if 50% point is too low and would cause rough running of the engine
- 90%&95% boiling point(Final boiling point) presents the amount of heavy fraction which is hard to evaporate. Higher final boiling point would enlarge the droplets in the injection jet, and has negative influence on the combustion process .

(90%及95%馏出温度表示难以蒸发的重馏分，偏高可导致喷雾蒸发困难，进而影响燃烧过程)

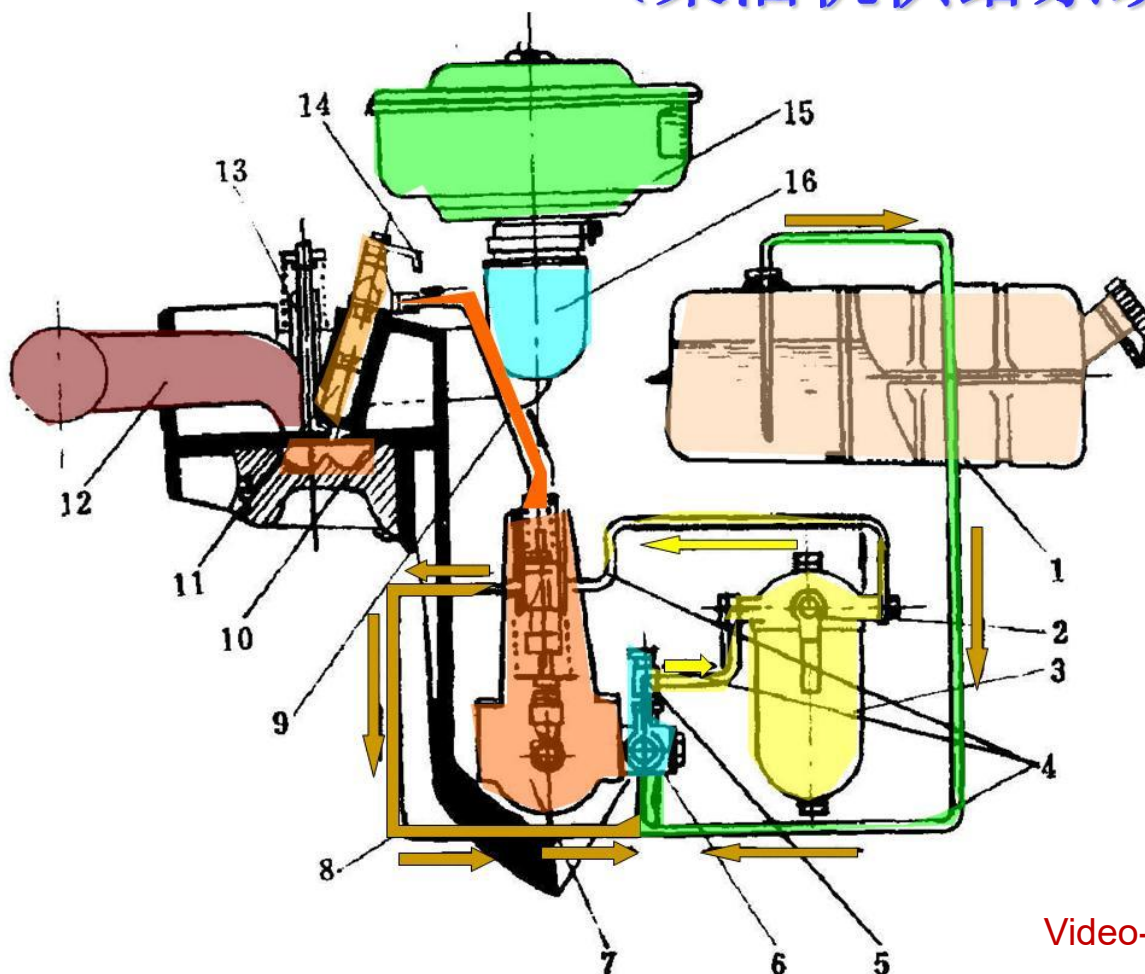
3. Atomization Quality(雾化性)—Viscosity(粘度)

- too low: insufficient lubrication of the fuel injection system
- Too high: larger droplet size, poorer mixture formation, lower performance, higher soot emissions
- kinematic viscosity(运动粘度): $(2.5 \sim 8) \times 10^{-6} \text{ m}^2 / \text{s}$ at 20°C

4. Flow ability(流动性)—Freezing Point(凝点)

- $T \downarrow$ --cloud point(浊点) —lose flow ability
- Diesel Brands: 10、5、0、-10、-20、-35 and -50

5.2 Composition of Fuel Supply System for Diesel Engine (柴油机供给系统组成)

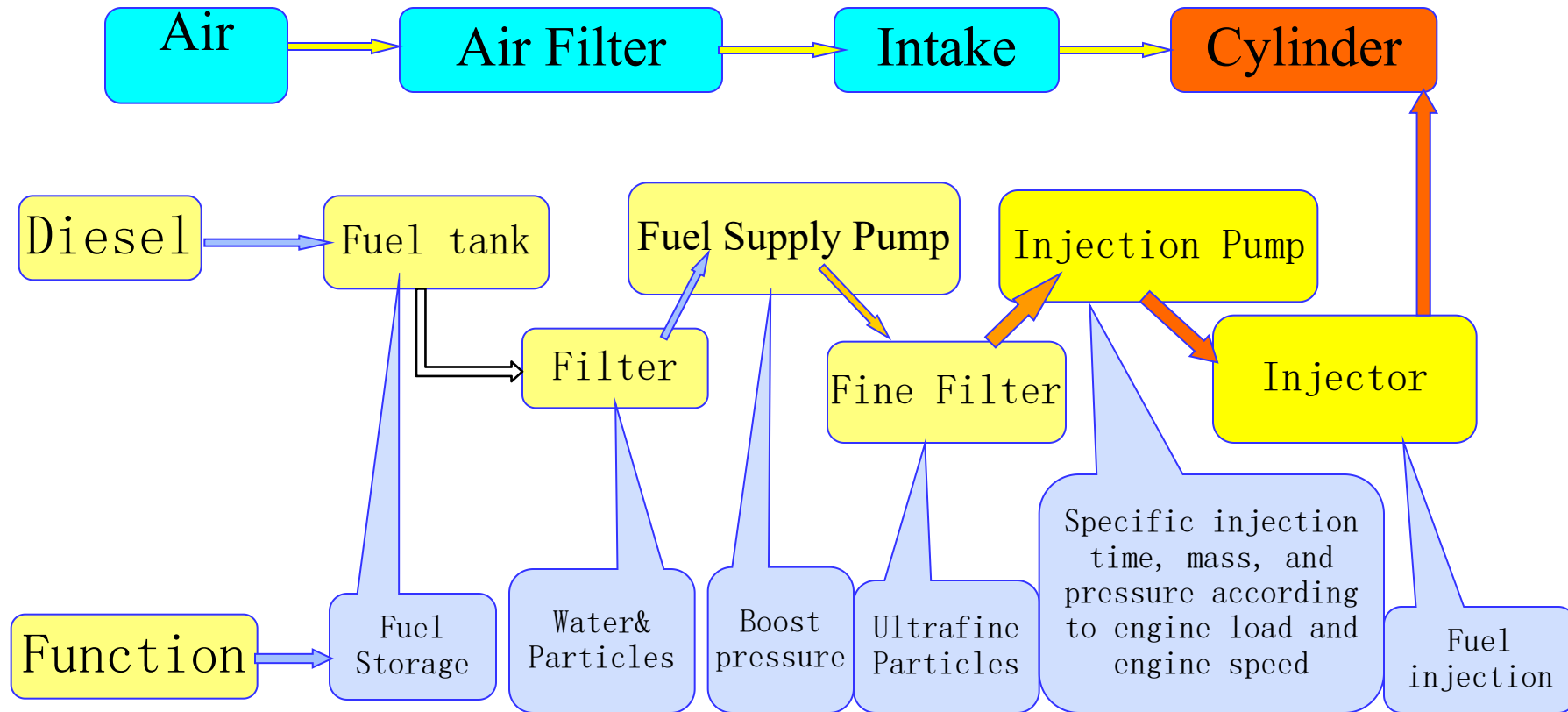


- 1- Fuel tank
- 2- Overflow valve
- 3- Fuel Filter
- 4- Low pressure pipe
- 5- Manual fuel supply pump
- 6- Auto fuel supply pump
- 7- Fuel injection pump
- 8- Fuel return pipe
- 9- High-pressure fuel pipe
- 10- Combustion Chamber
- 11- Fuel injector
- 12- Scavenge pipe
- 13- Exhaust valve
- 14- Fuel drain
- 15- Air filter
- 16- Intake manifold

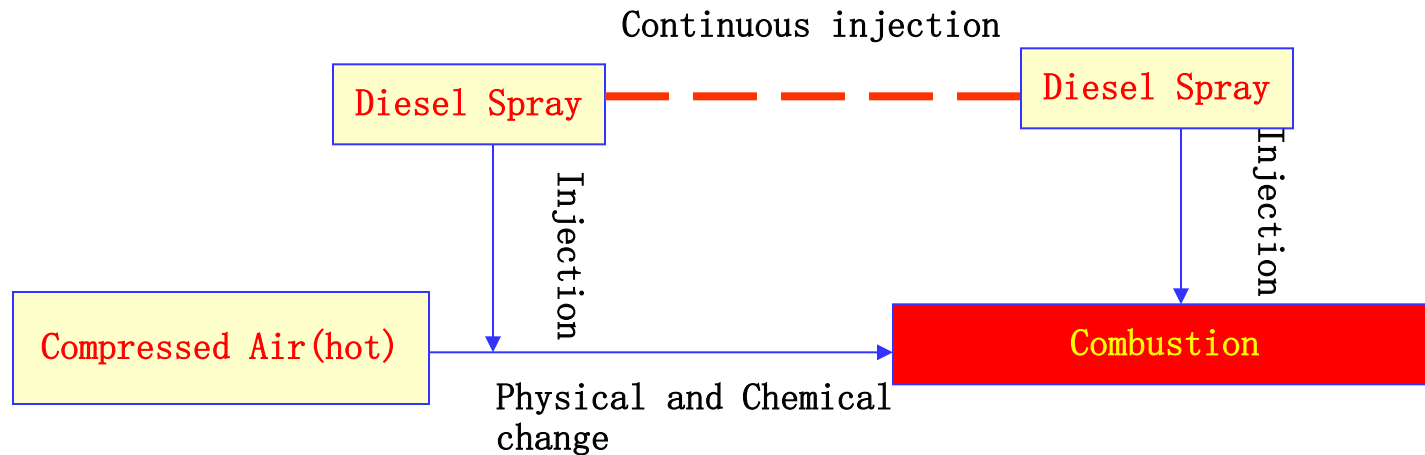
- 1-柴油箱;
- 2-溢油阀;
- 3-滤清器;
- 4-低压油管;
- 5-手动输油泵;
- 6-机动输油泵;
- 7-喷油泵;
- 8-回油管;
- 9-高压油管;
- 10-燃烧室;
- 11-喷油器;
- 12-排气管;
- 13-排气门;
- 14-泄油管;
- 15-空气滤清器
- 16-进气管;

Video-1

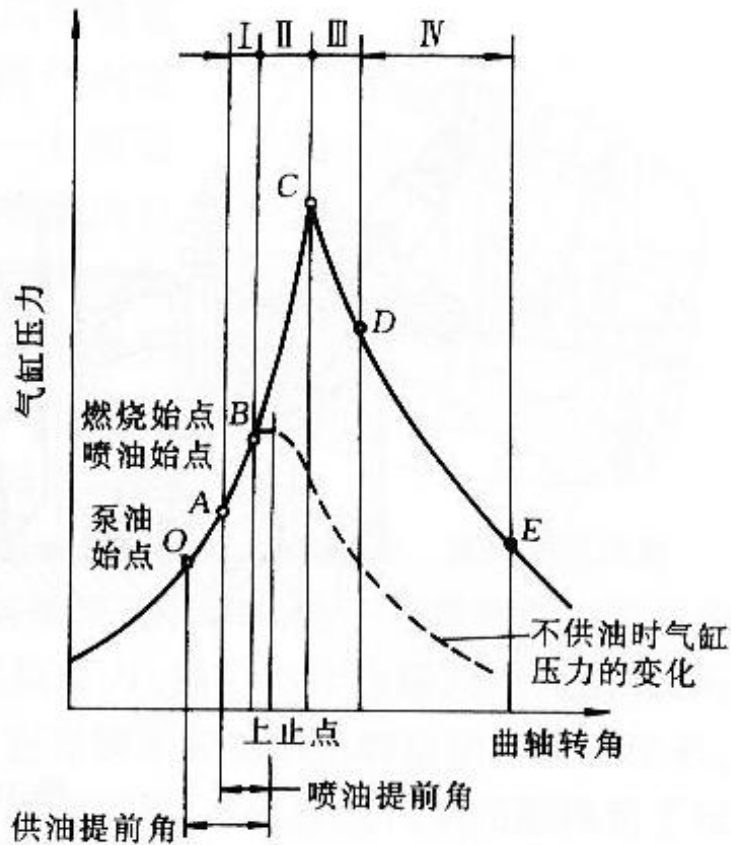
Function of Fuel Supply System



5.2.1 Mixture formation and Combustion Chamber (混合气形成及燃烧室)



- **Mixture formation: Atomization(空间雾化) and Evaporation(油膜蒸发)**
- **Combustion Characteristics:**
 - ✓ Little time and space for mixture formation and combustion(0.0017-0.004s, 15-35° CA)
 - ✓ High viscosity—hard to atomize and evaporate— high injection pressure
 - ✓ Zones with different O₂ concentrations—different emissions in different zones
 - ✓ Combustion process proceeds with fuel injection and mixture formation— excess air needed



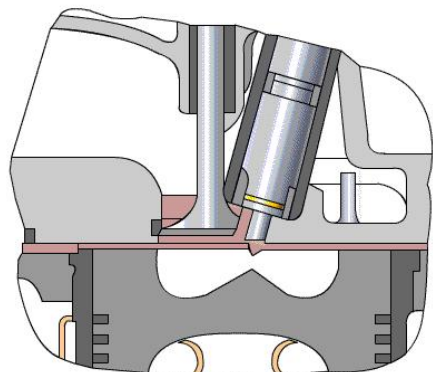
柴油机气缸压力与曲轴转角的关系

I—滞燃期； II—速燃期； III—缓燃期； IV—后燃期

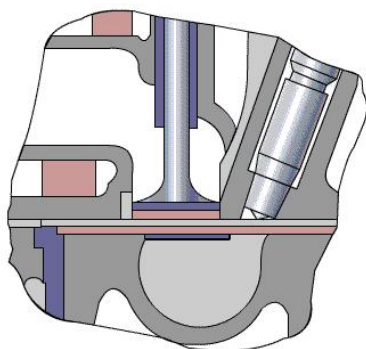
Combustion process can be divided into four periods:

- **A-B ignition delay period (滞燃期)**
From start of injection to start of combustion (combustion line leaves compression line)
- **B-C rapid combustion period (急燃期)**
From start of combustion to appearance of maximum cylinder pressure
- **C-D Slow combustion period (缓燃期)**
From appearance of maximum cylinder pressure to maximum cylinder temperature
- **D-E After combustion period (后燃期)**

5.2.1 Mixture formation and Combustion Chamber (混合气形成及燃烧室)



ω型燃烧室



球型燃烧室

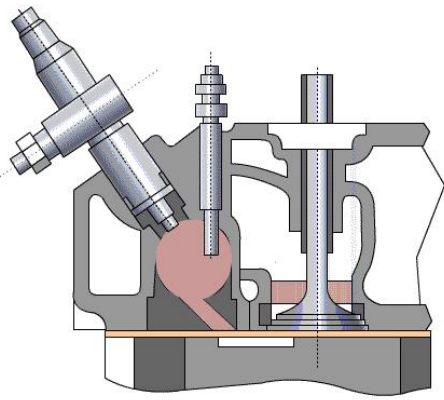
1、Open combustion Chamber
(统一式燃烧室)

{ ω Type (ω型)
Spherical (球型)

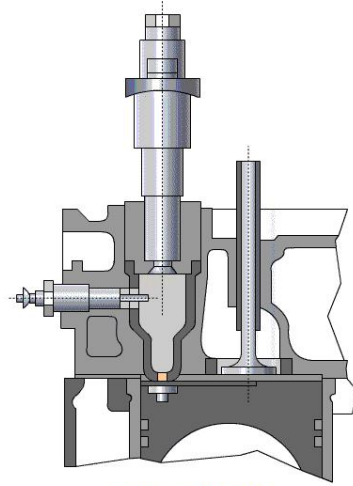
Open Combustion Chamber

(统一式燃烧室)

- The actual combustion chamber is formed by recess in the piston head and bottom of cylinder head
- Characteristics:
 - ✓ High combustion efficiency
 - ✓ Less heat loss
 - ✓ Requirements for higher injection pressure
 - ✓ Used in Direct Fuel injection Method
- 单纯由活塞顶的凹面与气缸盖下底面所形成的燃烧空间这种形式的燃烧室燃烧效率高，热损失较少；但要求喷油压力高（17~22MPa），油品质量较好。一般分为球型燃烧室及ω型燃烧室。



涡流室式燃烧室



预燃室式燃烧室

Divided Combustion Chamber

(分隔式燃烧室)

- Combustion chamber consists of a main combustion chamber and a secondary combustion chamber.
- Main combustion chamber is located in piston crown, while secondary combustion chamber is located in cylinder head

2、Divided Combustion Chamber (分隔式燃烧室)

Whirl combustion chamber

涡流室燃烧室

Pre-combustion chamber

预燃式燃烧室

Fuel is injected into secondary combustion chamber first, and a pre-combustion process is carried out. After that, the mixture is pushed back into the main combustion chamber to finish the whole combustion process

• 分隔式燃烧室一般由两部分组成，一部分位于活塞顶与缸盖底面之间，称为主燃烧室；另一部分位于气缸盖中，称为副燃烧室。

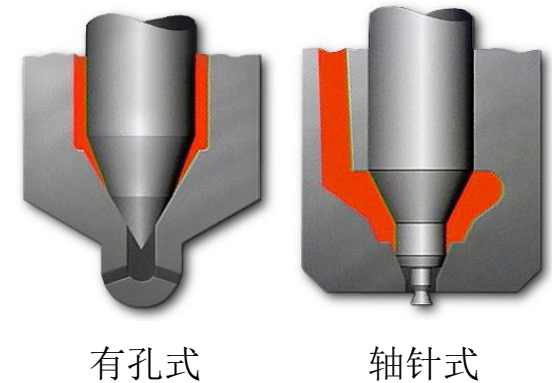
• 喷油器将燃油喷入副燃烧室，经预燃后混合气反冲至主燃烧室完成燃烧过程。热损失偏高，但对喷油压力要求较低

Characteristics:

- ✓ Higher heat loss
- ✓ Less heat loss
- ✓ Less requirement of high injection pressure
- ✓ Used in indirect fuel injection method

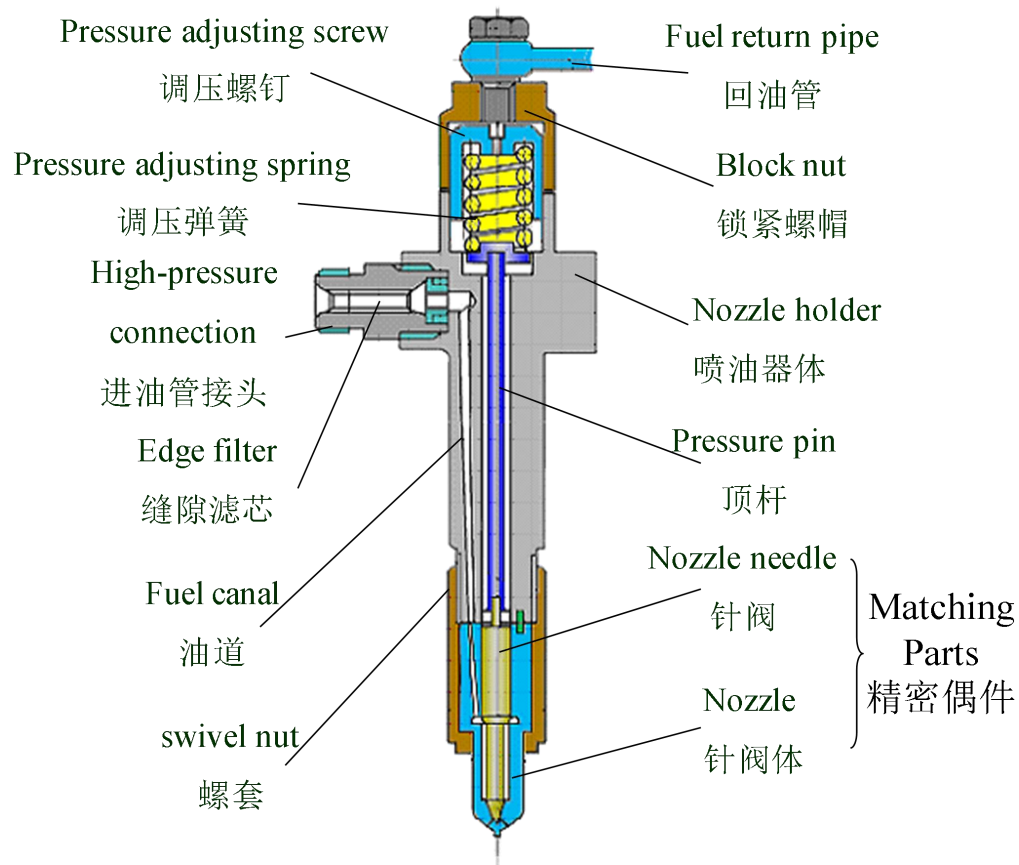
5.3 Fuel Injector (喷油器)

- **Function:** Inject the fuel provided by injection pump into the cylinder with fine atomization and distribution
- **Basic design: pintle nozzles and hole-type nozzles**
- **Requirements:**
 - ✓ Appropriate spray corn angle (喷雾锥角) with fine atomization
 - ✓ Injection with specific pressure and spray penetration(贯穿距)
 - ✓ Quick cut-off without droplet leakage
 - ✓ Appropriate spray shape and direction matched with combustion chamber



- 1、功用：将喷油泵供给的高压柴油，以一定的压力，呈雾状喷入燃烧室。
- 2、分类：有孔式和轴针式两种。
 - ①合适的喷注锥角，良好的雾化质量；
- 3、要求：
 - ②有一定的喷射压力和射程；
 - ③喷油终了迅速停油，无滴漏现象；
 - ④油束形状与方向，适应燃烧室。

Structure of Injectors

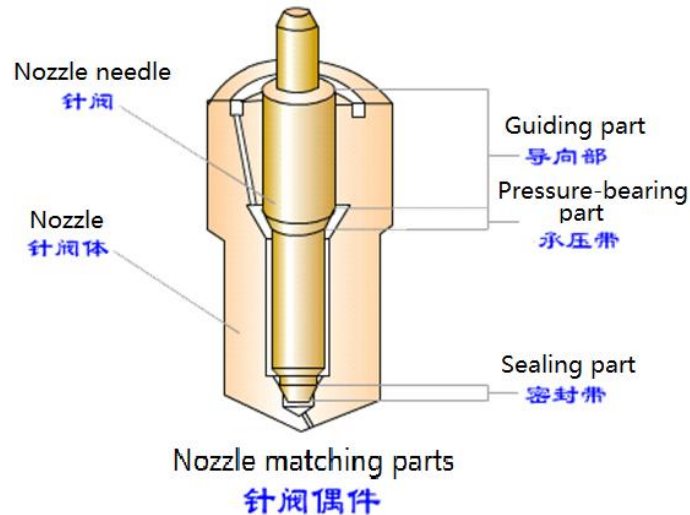


Structure of hole-type nozzle

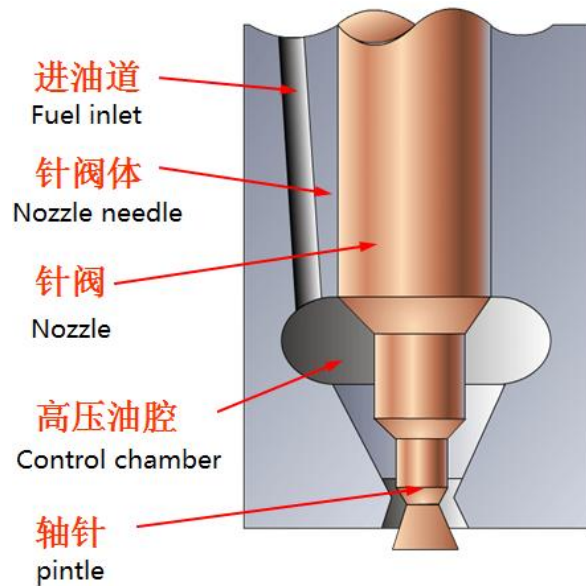
Video-2

Working principle

- **Injection:** fuel in high-pressure chamber forces the nozzle needle to move upwards → sealing cone departs from seat → injection
- **喷油:** 喷油泵泵油 → 环形高压油腔进油 → (承压锥面承受油压推力) 针阀上行 → 密封锥面离座 → 喷孔喷油。
- **Cut-off:** Fuel pressure drops → pressure adjusting spring forces the needle to move downwards → sealing cone back → nozzle closed by needle
- **停喷:** 喷油泵停止泵油 → 高压油腔油压下降 → 调压弹簧伸张 → 顶杆推针阀下行 → 密封锥面落座 → 喷孔关闭 → 不喷油。
- **Fuel return:** Part of high-pressure fuel returns to filter through gap between needle valve couple, lubricating and cooling the needle valve couple
- **回油:** 少量高压油经针阀偶件间间隙挤上, 经回油管流回柴油箱或柴油滤清器, 润滑、冷却针阀偶件。



- **Hole-type nozzle:**
- Position and direction of the nozzle matches with the geometry of combustion chamber
- Higher injection pressure
- Higher machining accuracy



- **Pintle nozzle**
- Without sac volume
- Larger nozzle diameter, easy machining, lower cost
- No droplet leakage during cut-off process
- Harder to control injection mass precisely

5.4 High-pressure Pump (喷油泵)

• Function

Deliver high-pressure fuel to injector according to engine needs, including injection order, engine load, injection timing, injection pressure, and injection mass

(按照发动机的工作顺序，负荷大小，定时、定量、定压地向喷油器输送高压柴油。)

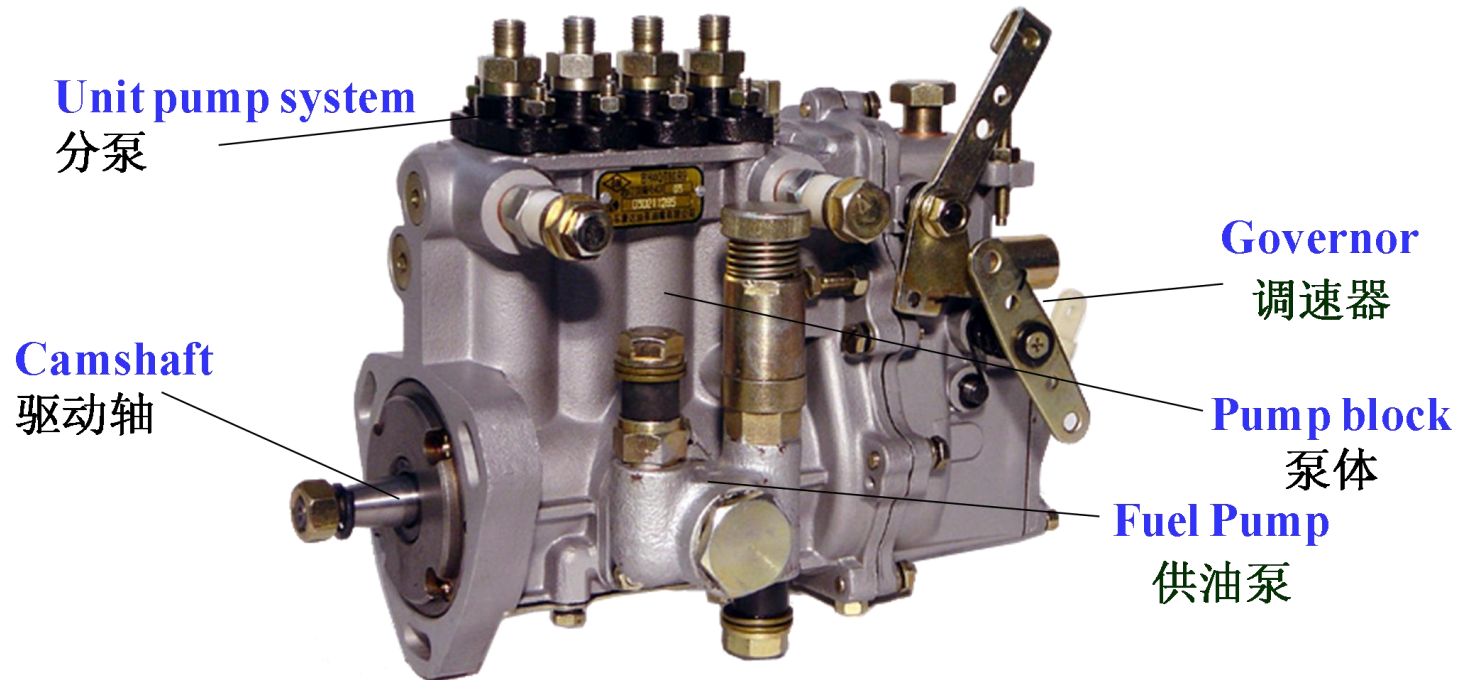
• Classification

- ✓ Piston pump
- ✓ Distributor Injection pump
- ✓ Pump-Nozzle System
- ① 柱塞式喷油泵 (A、B、P、Z型)
- ② 转子分配式喷油泵 (VE)
- ③ 喷油泵- 喷油器 (单体泵或泵喷嘴)

Requirements for multi-cylinder fuel pump

- ✓ Same amount of injection mass for each cylinder (difference within 3%~4%)
- ✓ Same injection advance angle (difference within 0.5~1 deg CA), with angle automatically changing with engine load and speed
- ✓ Same injection duration
- ✓ Swift cut-off process without droplet leakage

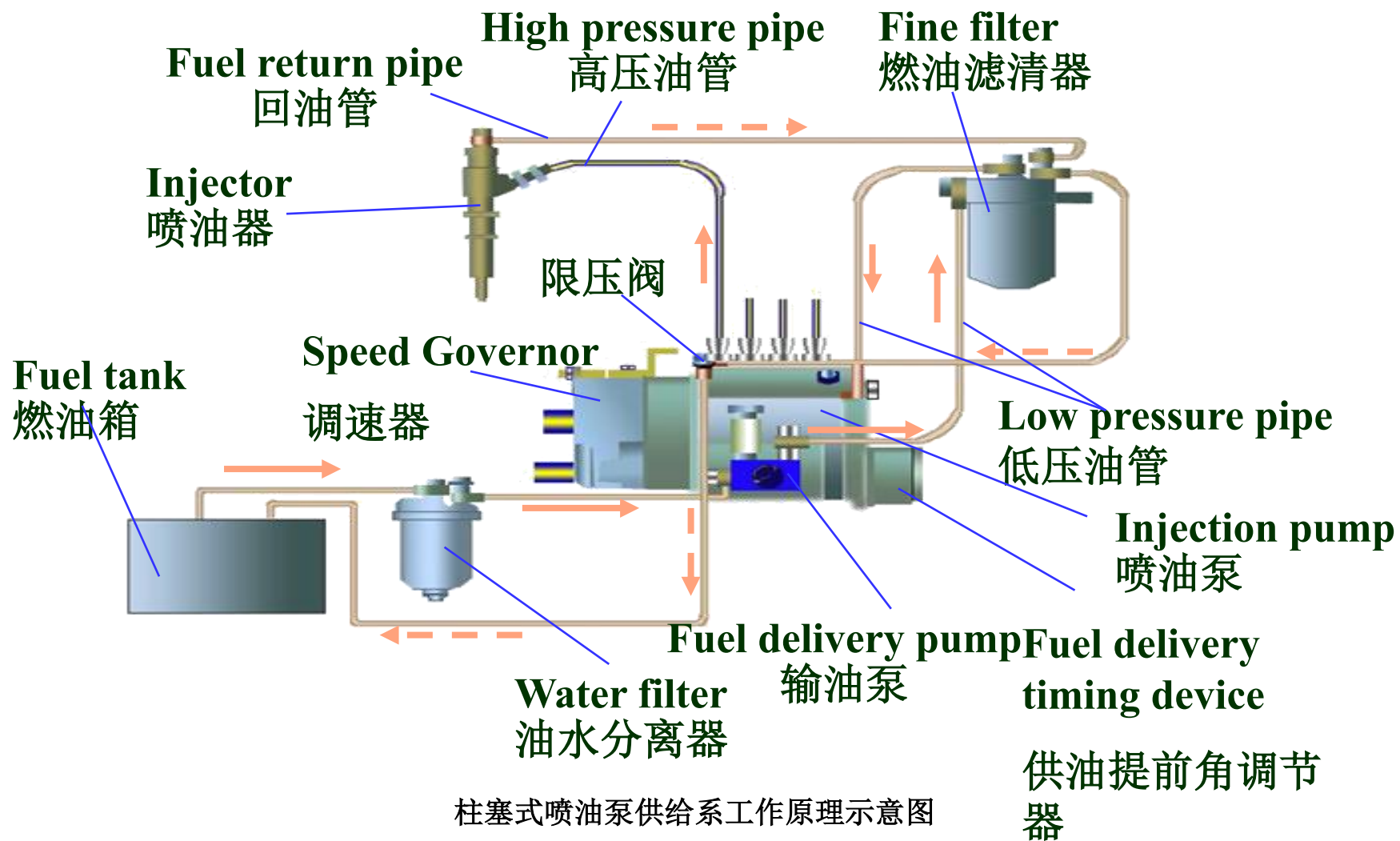
5.4 In-line Fuel injection Pumps (直列柱塞式喷油泵)



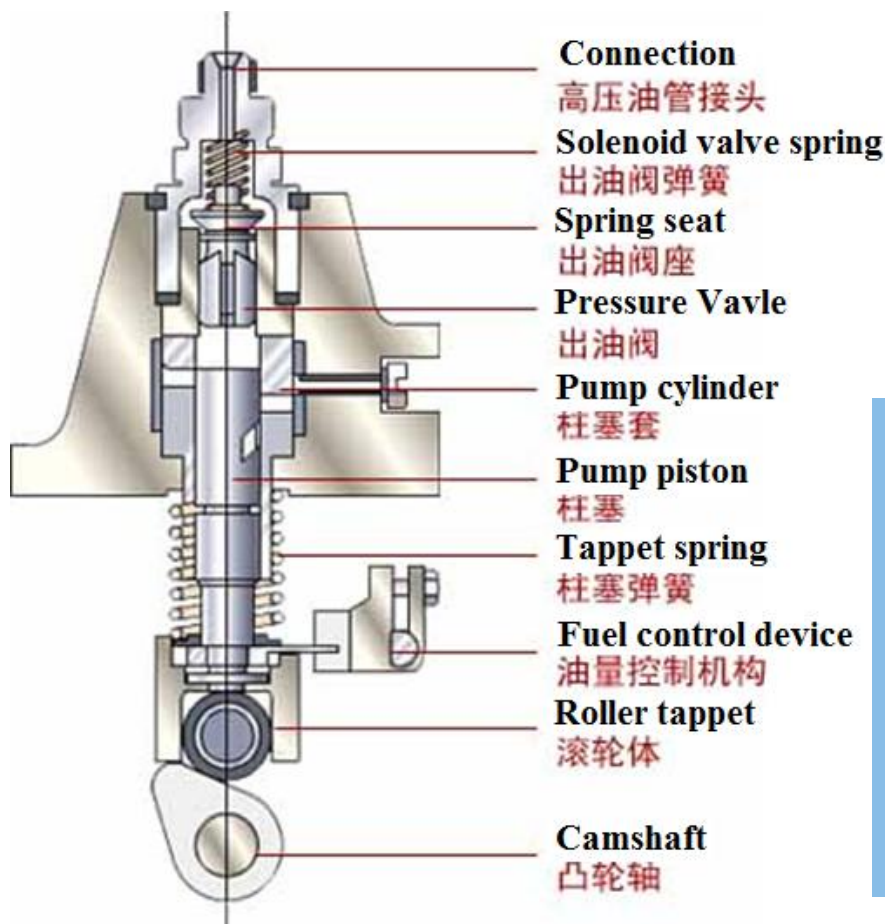
- **Pumping mechanism**
(泵油机构)
- **Fuel control mechanism**
(油量调节机构)

- **Driving mechanism**
(驱动机构)
- **Pump block**
(油泵体)

Video-3

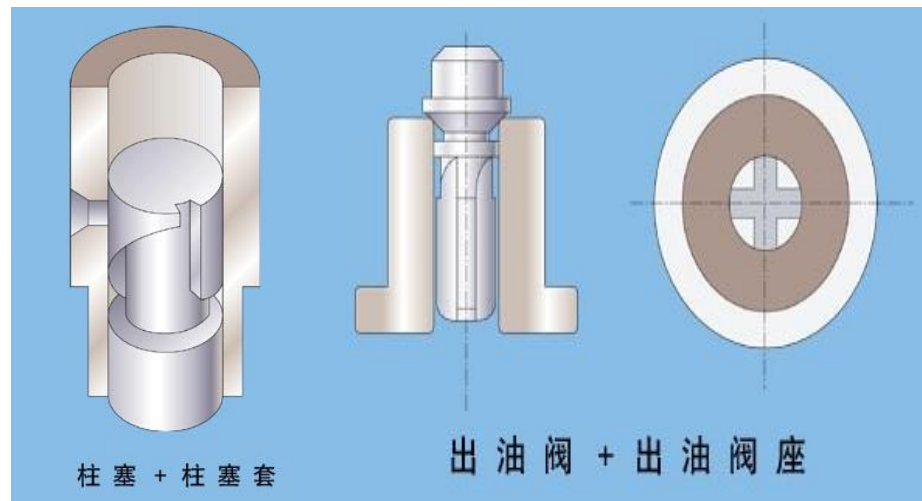


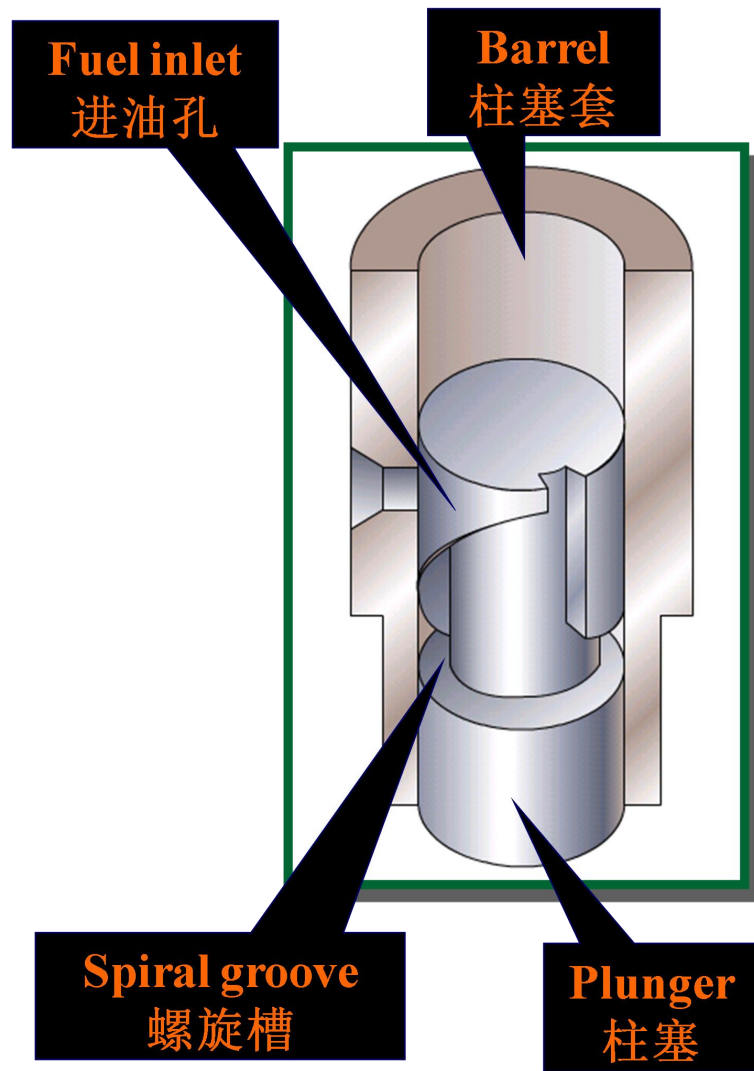
5.4.1 Pumping Mechanism (泵油机构)



Components:

- ✓ Plunger and barrel assembly 柱塞偶件
- ✓ Solenoid Valve Couple 出油阀偶件
- ✓ Roller assembly 滚轮体组





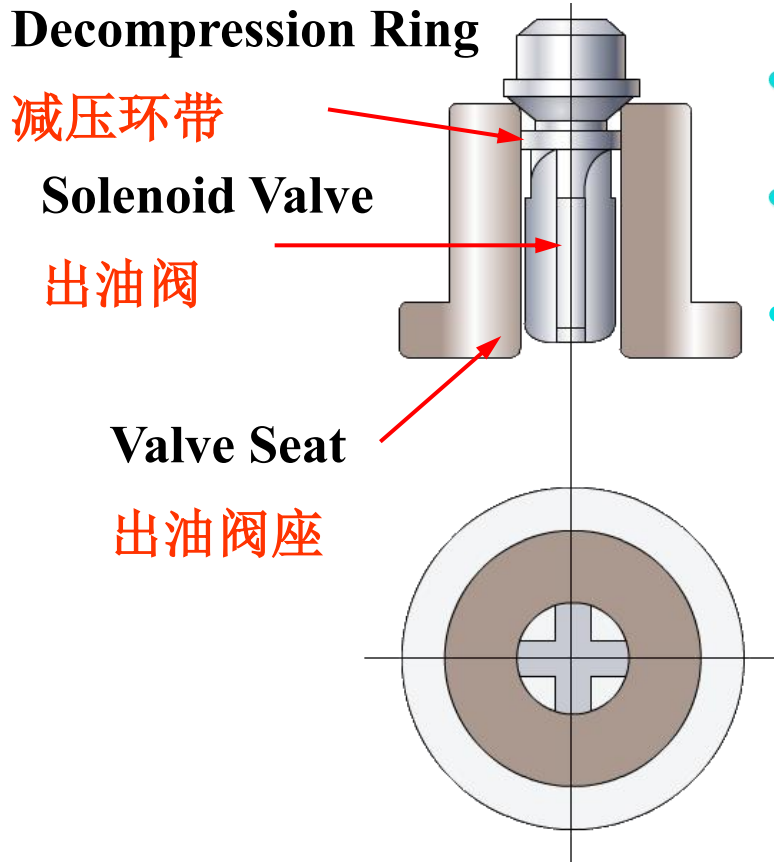
- **Plunger and barrel assembly**

- ✓ A pair of precise coupled device, not interchangeable after grinding with each other
- ✓ High precision and fineness with fit clearance of 0.001~0.003mm
- ✓ Head of plunger is machined with spiral groove to control injection mass

- **组成：**柱塞及柱塞套

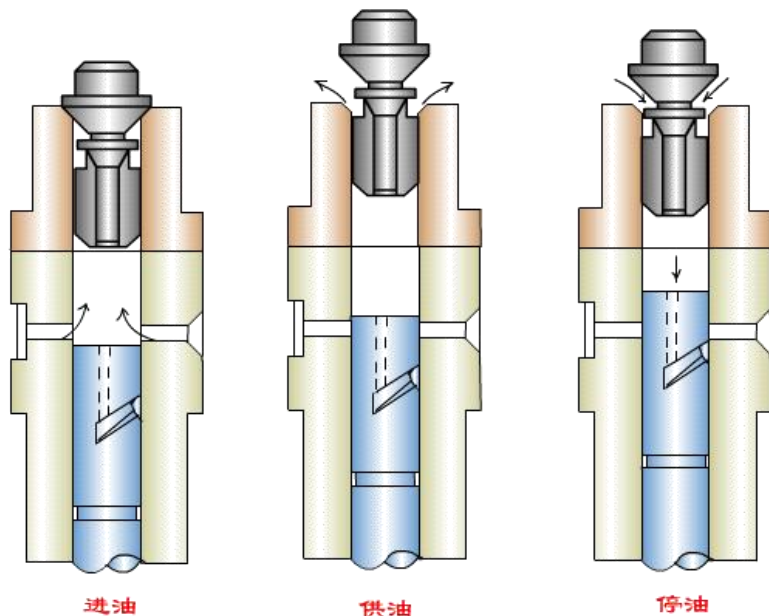
- ▶ **柱塞和柱塞套**是一对**精密偶件**，经配对研磨后不能互换，要求有高的精度和光洁度和好的耐磨性，其配合间隙为0.001~0.003mm。
- ▶ 柱塞头部圆柱面上切有**斜槽**，并通过径向孔、轴向孔与顶部相通，其目的是改变循环供油量

5.4.2 Solenoid Valve Couple (出油阀偶件)



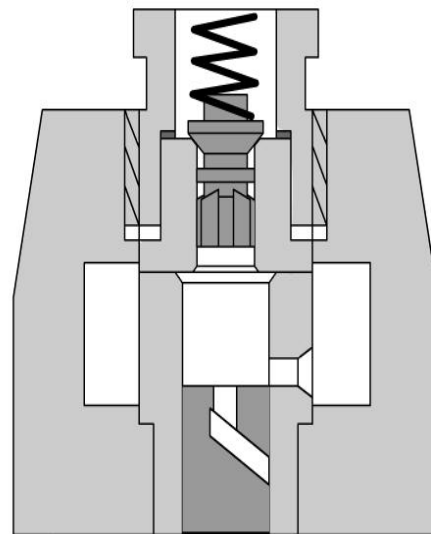
- **Solenoid valve couple:** Another pair of precise couple device with fit clearance of 0.01mm
- **Component:** Solenoid valve seat, solenoid valve spring, decompression ring
- **Solenoid valve:** A one-way valve matches with cone of valve seat under pressure
- The bottom part of Solenoid valve is a cross section, which could deliver diesel and guide Decompression ring is located under the valve cone.
- **组成:** 出油阀及座、出油阀弹簧、减压环带等。
- ▶ **出油阀和出油阀座**也是一对**精密偶件**, 配对研磨后不能互换, 其配合间隙为0.01。
- ▶ **出油阀**是一个单向阀, 在弹簧压力作用下, 阀上部圆锥面与阀座严密配合。
- ▶ 出油阀下部呈十字断面, 既能导向, 又能通过柴油。出油阀的锥面下有一个小的圆柱面, 称为减压环带。

Working Principle



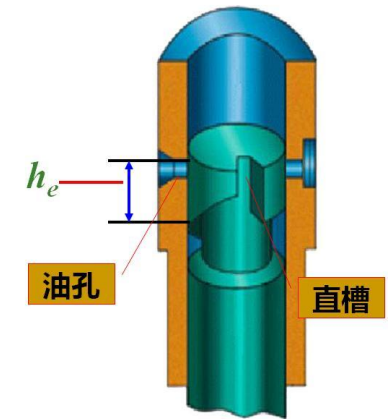
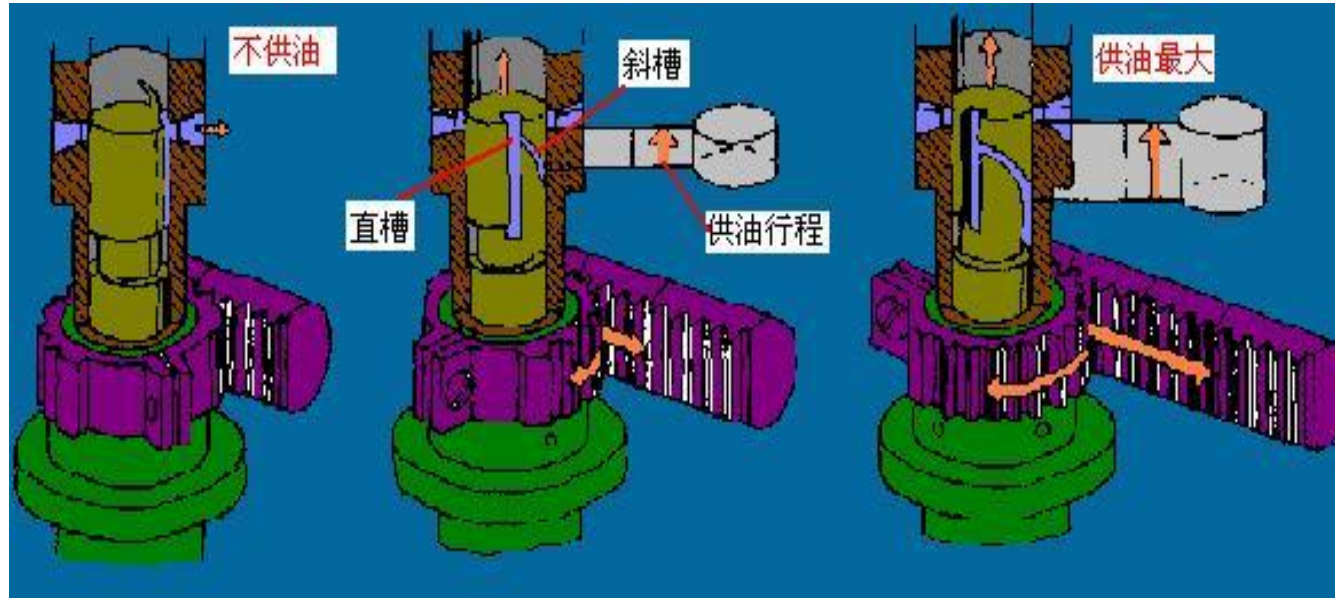
柱塞泵的泵油过程

- Fuel inlet process
- Fuel delivery process
- Fuel return process



开始

5.4.3 Fuel Control Mechanism (油量调节机构)



- **Function:** Change the effective stroke h_e by rotating the plunger according to different engine speed and load. A small plunger lift corresponds to an early start of delivery; a large plunger lift corresponds to a late start of delivery.

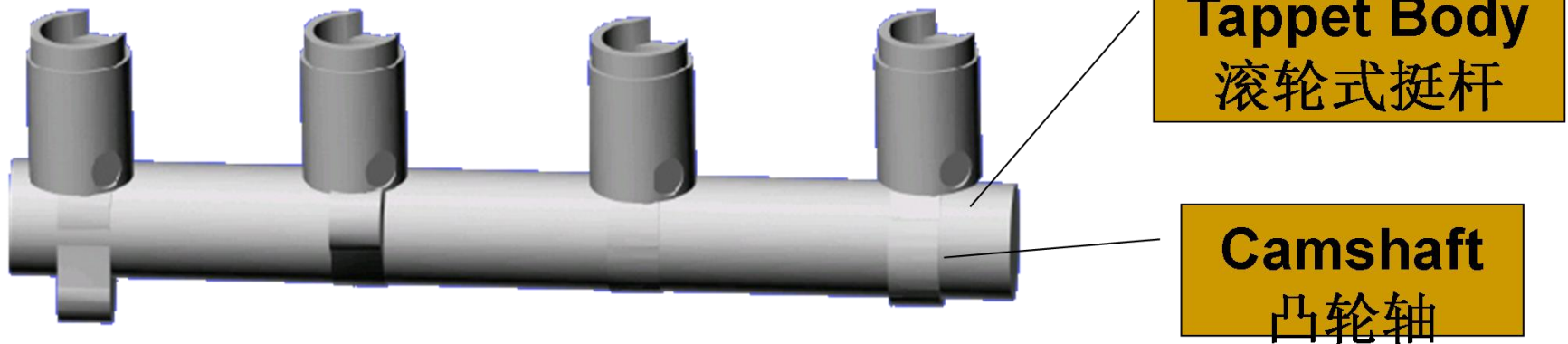
5.4.4 Driving mechanism (驱动机构)

- **Component**

{ Roller part: Roller tappet and tappet roller
Camshaft

- **Structure**

Camshaft is supported by bearings at the end of pump and joints with shaft joint, the number of cam equals cylinder number, and the rotation speed of pump camshaft is half the rotation speed of crankshaft.



5.4.5 Pump block (喷油泵体)

- **Function:** Provide a mounting foundation for fuel pump, fuel control, and driving mechanism
- **Structure:** Pump block of A-type pump is integrated with window on the side. This type of design is easier for dismounting and adjustment. However, rigidity is less which leads to lower maximum pumping pressure.
- **Requirement:** Sufficient intensity, rigidity and fine tightness, easy for dismounting and repair.

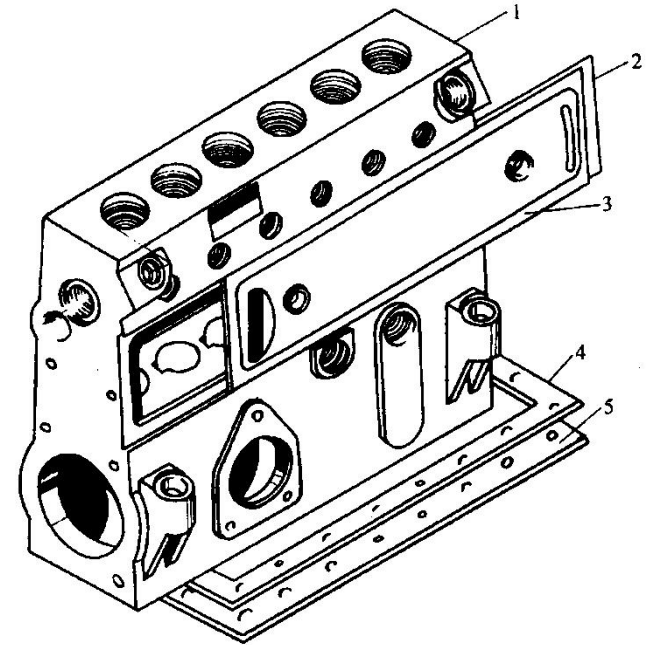


图 5-18 A 型喷油泵泵体

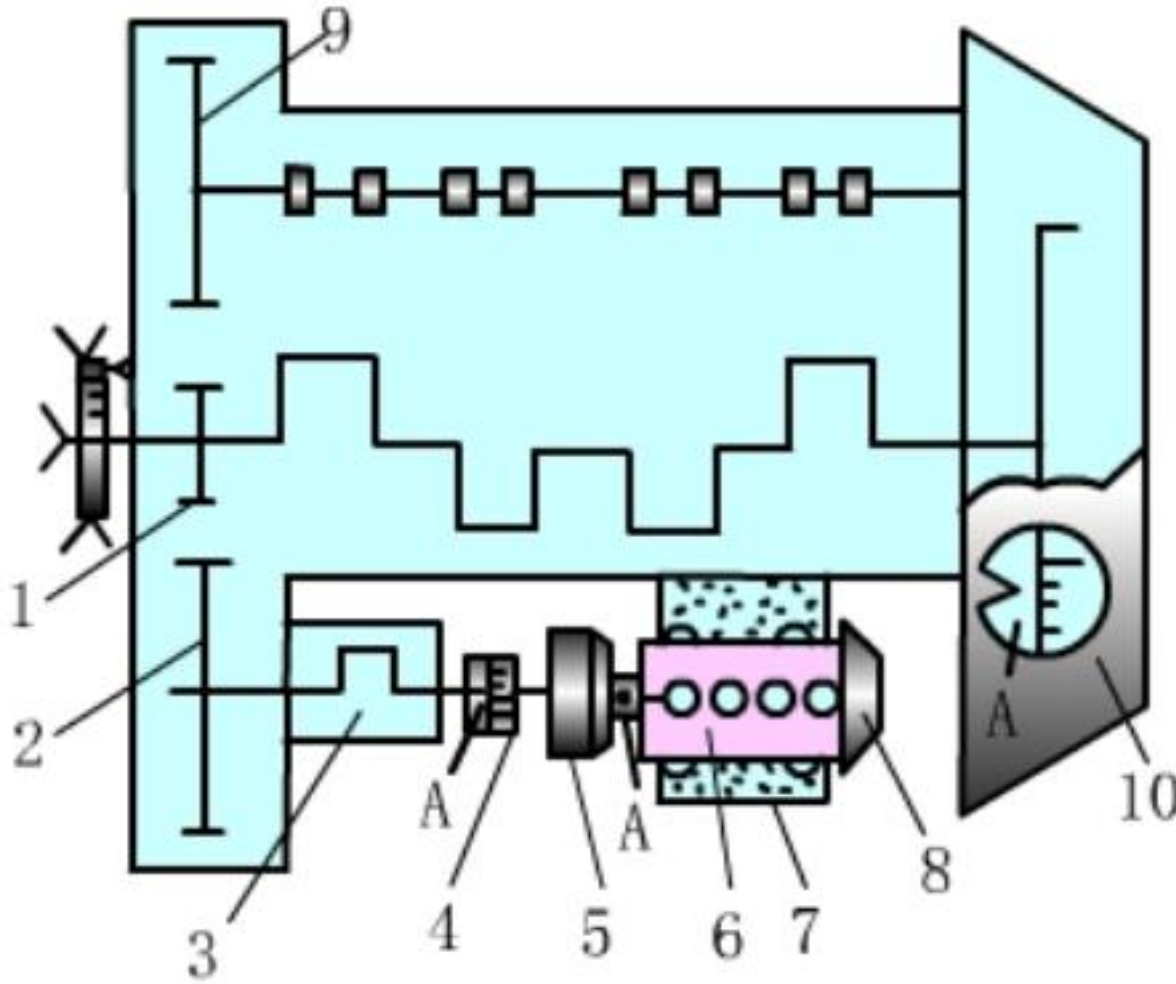
1—喷油泵体 2、4—衬垫 3—侧盖 5—底盖

功用: 是泵油机构、供油量调节机构、驱动机构的安装基体。

结构: A型泵泵体是整体式，侧面开窗，便于拆装、调整，但泵体刚性差，允许最大泵端油压较低。泵体上有放气螺钉。泵体下端的凸轮室装有传动机构的润滑油。

要求: 有足够的强度、刚度和良好的密封性；便于拆装、调整和维修。

Layout of Fuel Delivery Mechanism



1-Crankshaft timing gear

曲轴正时齿轮

2-Injection pump driving gear

喷油泵驱动齿轮

3-Air compressor crankshaft

空压机曲轴（制动）

4-Joint 关节器

5-Fuel delivery timing device

供油提前角自动调节器

6-Injection pump 喷油泵

7-Layer board 托板

8-Speed governor 调速器

9-Camshaft timing gear

凸轮轴正时齿轮

10-Injection timing sign(flywheel)

飞轮上的喷油正时标记

5.4.6 Timing Device

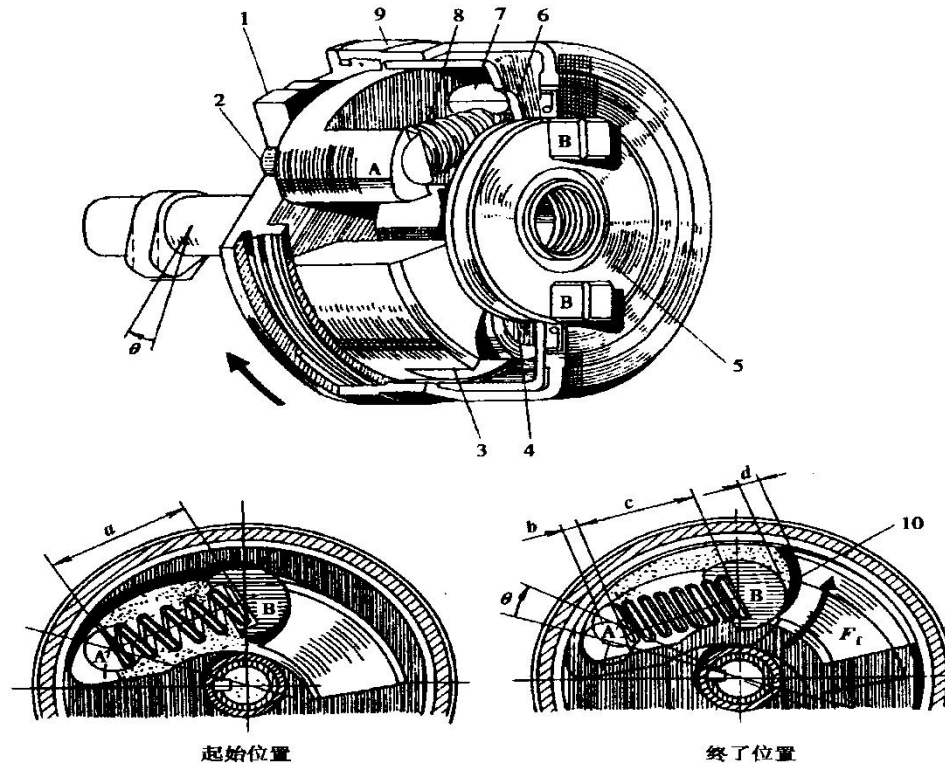


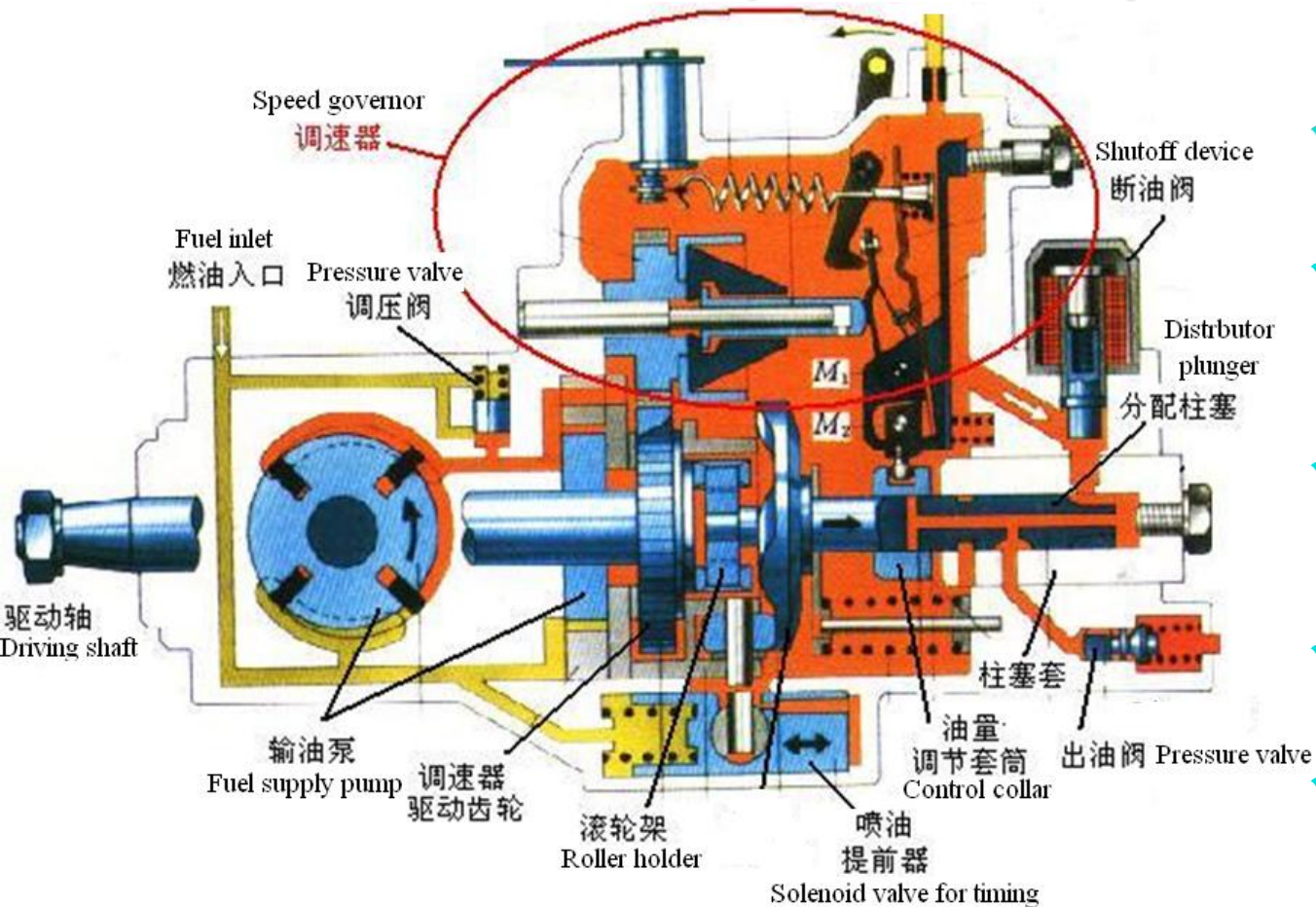
图 6-23 机械式自动喷油提前器

1-从动盘;2-飞锤销;3-飞锤;4、7-传动销;5-主动盘凸缘;6-主动盘;8-提前器弹簧;9-防护罩;10-飞锤圆弧面

a -起始时弹簧长度; b -终了时飞锤销的移动距离; c -终了时弹簧长度; d -终了时飞锤移动距离; θ -提前角调节范围; F_t -飞锤离心力

- Engine speed $\uparrow \rightarrow$ Injection timing $\uparrow \rightarrow$ Fuel delivery timing $\uparrow$

5.5 Distributor Injection Pump (分配式油泵)



Component:

- ✓ Driving mechanism
驱动机构
- ✓ Low-pressure supply pump
低压输油泵
- ✓ High-pressure supply pump
高压输油泵
- ✓ Timing device unit
喷油时刻调整装置
- ✓ Speed and fuel governor
调速器



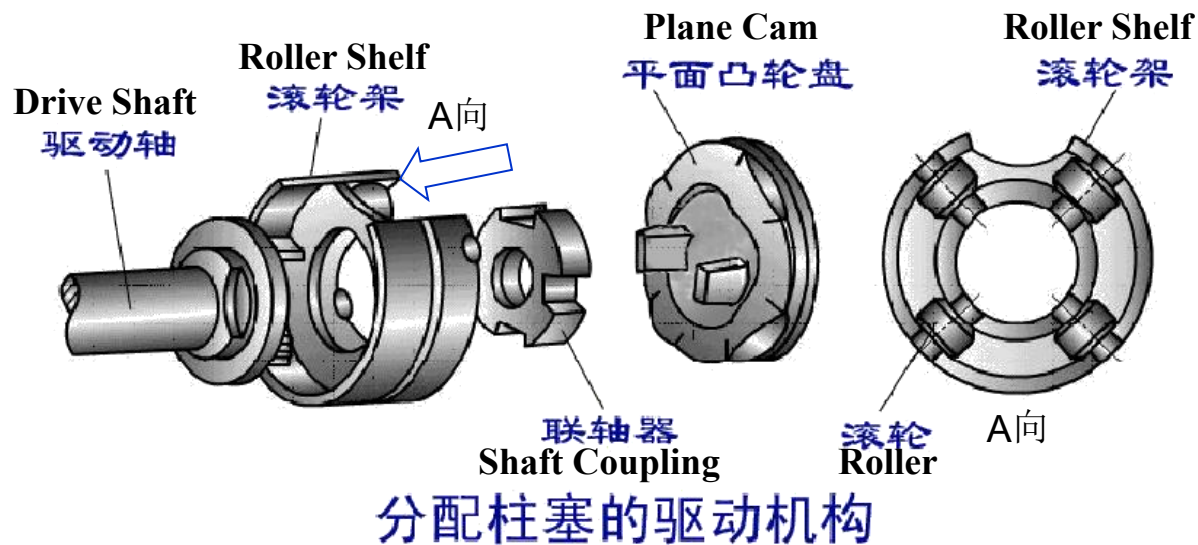
1—燃油箱 2—油水分离器 3—一级输油泵 4—二级输油泵 5—燃油滤清器 6—调压阀
7—分配式喷油泵传动轴 8—调速手柄 9—分配式喷油泵体 10—喷油器 11—回油管
12—分配式喷油泵 13—喷油提前器 14—调速器传动齿轮

- **Pump characteristics**

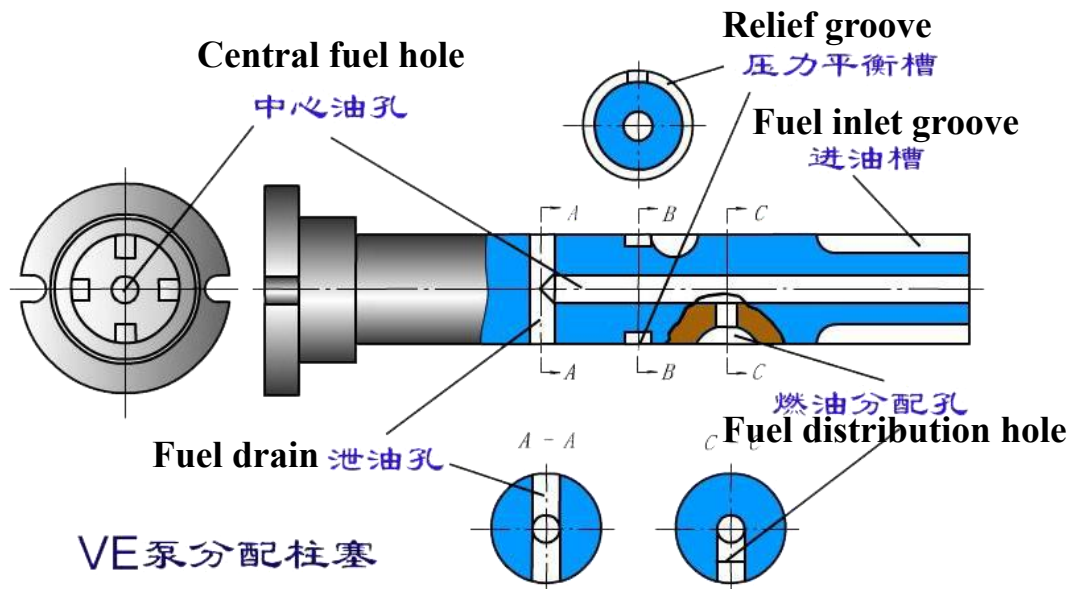
- One pump element is required for all engine cylinders because the pump plunger rotates at the camshaft speed
- The pump plunger distributes the fuel to the engine cylinders corresponding to the firing sequence
- Applied for high-speed engines for passenger cars up to six cylinders
- Suitable for engines with a large speed range
- Valve gear is exclusively lubricated by the fuel in contrast to individual injection system and in-line fuel injection pump

- **特点:**

- 通过一个泵体向各缸供油，柱塞转动速度与凸轮轴转速相同
- 柱塞按照各缸点火顺序向缸内分配燃油
- 应用于乘用车，适用高速柴油机
- 适用于转速范围较大的发动机
- 润滑仅通过回流的柴油完成（直列泵通过机油完成润滑）



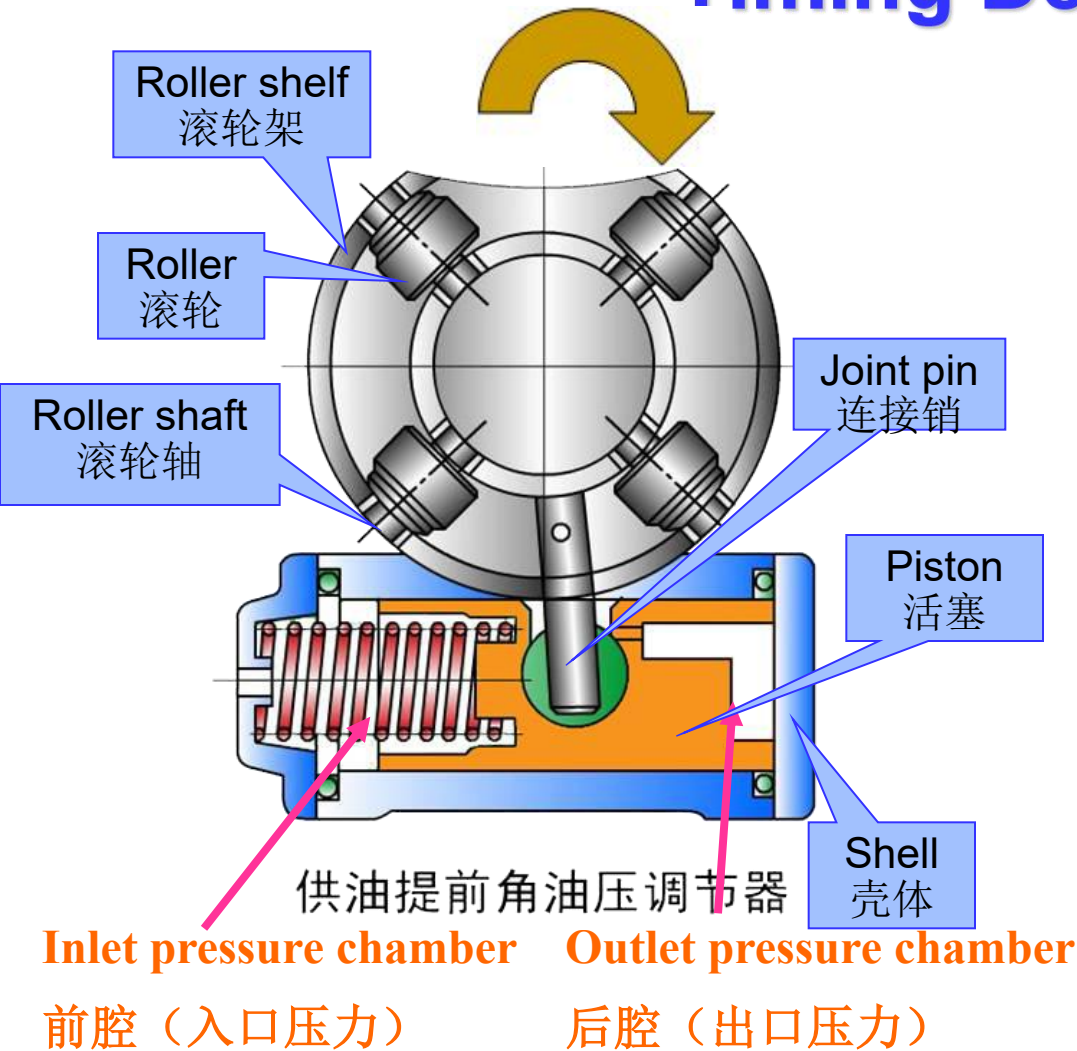
- Plane Cam is force to do axial reciprocating movement while rotated by drive shaft
- 驱动轴转动时，带动联轴器、端面凸轮盘同方向旋转，由于端面凸轮盘被柱塞复位弹簧压紧在滚轮架上，因此，端面凸轮迫使滚轮自转，并使端面凸轮盘作轴向往复直线运动。



- Distribution plunger also does rotational and reciprocating movement at the same time driven by plane cam
- Reciprocating movement for fuel pumping
- Rotational movement for fuel distribution

平面凸轮盘通过传动销镶嵌在分配柱塞圆盘端开口槽内，由于柱塞复位弹簧将柱塞压紧在端面凸轮盘上，因此带动分配柱塞旋转，凸轮型面又使分配柱塞在旋转的同时，还作往复直线运动。

Timing Device

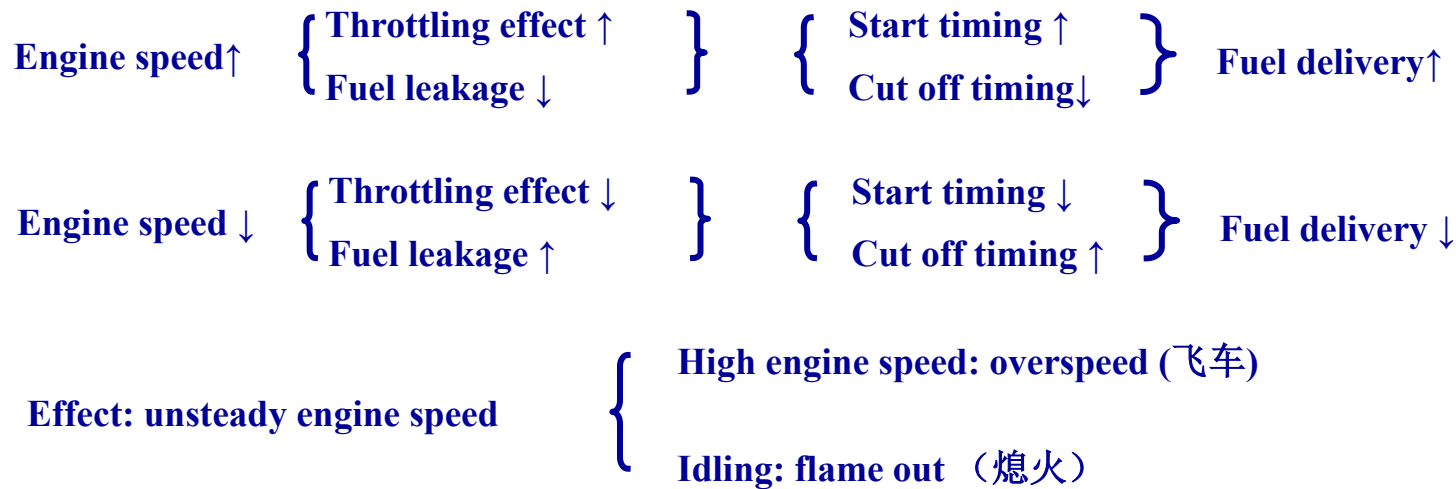


- Engine speeds up
 - Outlet pressure rises
 - Piston moves left
 - Joint pin leads roller shelf to rotate clockwise
 - Fuel delivery timing is in advance

发动机转速稳定时，作用在活塞两端的作用力相等，活塞平衡在某一位置。若转速升高，二级滑片式输油泵的出口压力增大，活塞左移，通过连接销3和传力销4带动滚轮架7绕其轴线转动一定的角度，直至活塞两端的作用力重新达到平衡，其旋转方向与平面凸轮盘的旋转方向相反，供油提前。

5.6 Speed Governor (调速器)

- **Function:** Stabilize engine speed under different engine loads by regulating fuel delivery
调速器是根据发动机负荷变化而自动调节供油量，从而保证发动机的转速在很小的范围内变化。
- **Speed performance of delivery pump:**



5.6 Speed Governor (调速器)

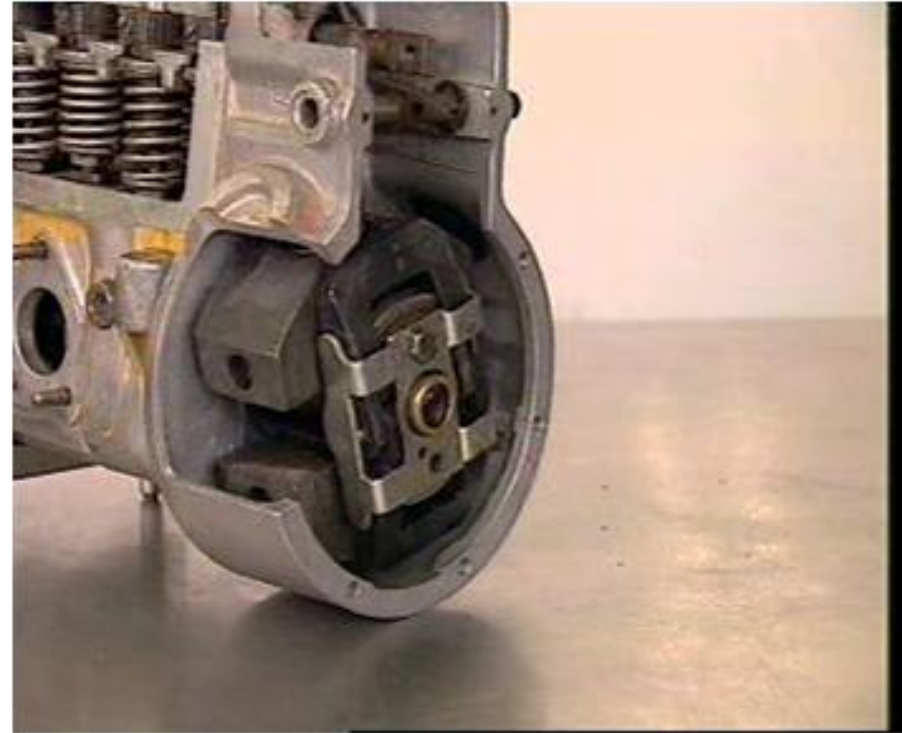
- **Classification:**

- ✓ In terms of function: two-speed governor, full-speed governor, fixed-speed governor, combined governor

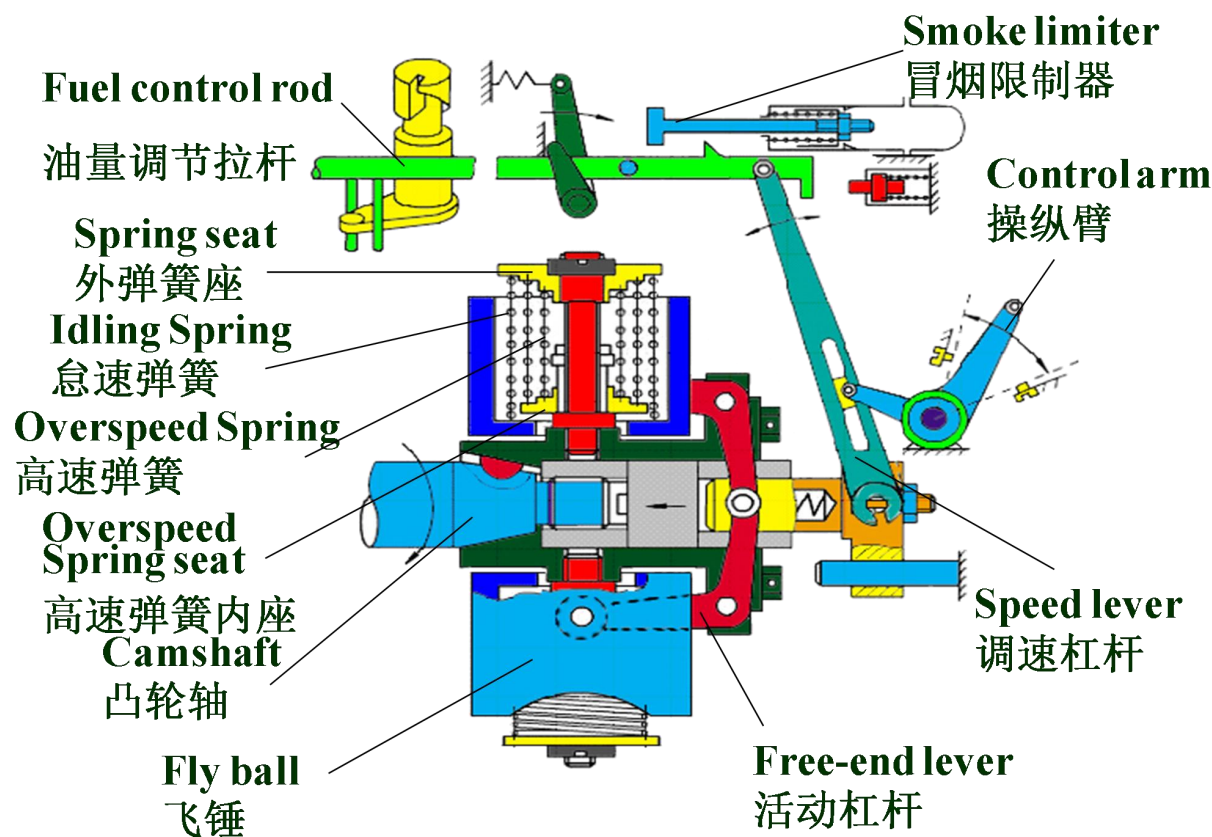
按功能分有两极调速器、全速调速器、定速调速器和综合调速器。

- ✓ In terms of sensor: pneumatic governor, centrifugal governor, complex governor

按转速传感分有气动式调速器、机械离心式调速器和复合式调速器。



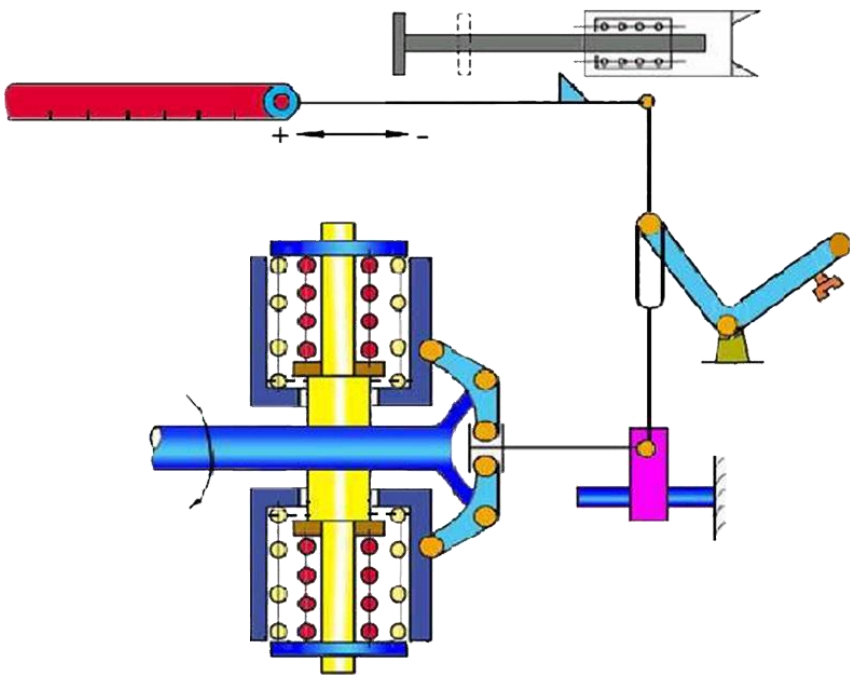
5.6.1 Two-speed governor (两极式调速器)



RQ型两极调速器

- **Function**
Stabilize and limit engine speed when it reaches its highest and lowest level
- Not function when engine speed is within the normal speed range

自动稳定和限制柴油机最低与最高转速，而在中间转速范围内则由驾驶员控制。



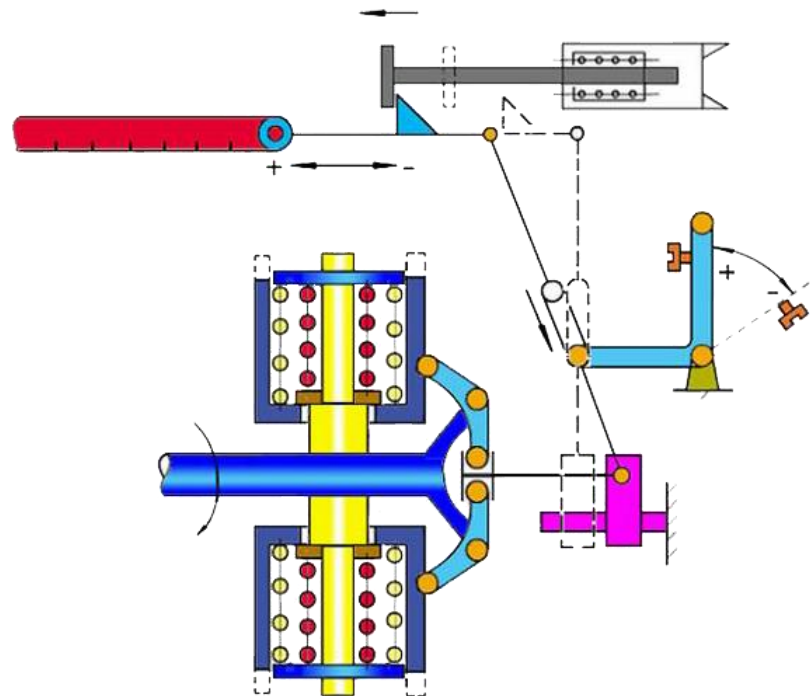
Stabilize idle speed:

- **Idle speed higher:**

Fly balls move outwards and pull the fuel control rod back to decrease fuel delivery

- **Idle speed lower:**

Fly balls move back and push the rod forward to increase fuel delivery



- **Engine speed under rated speed:**

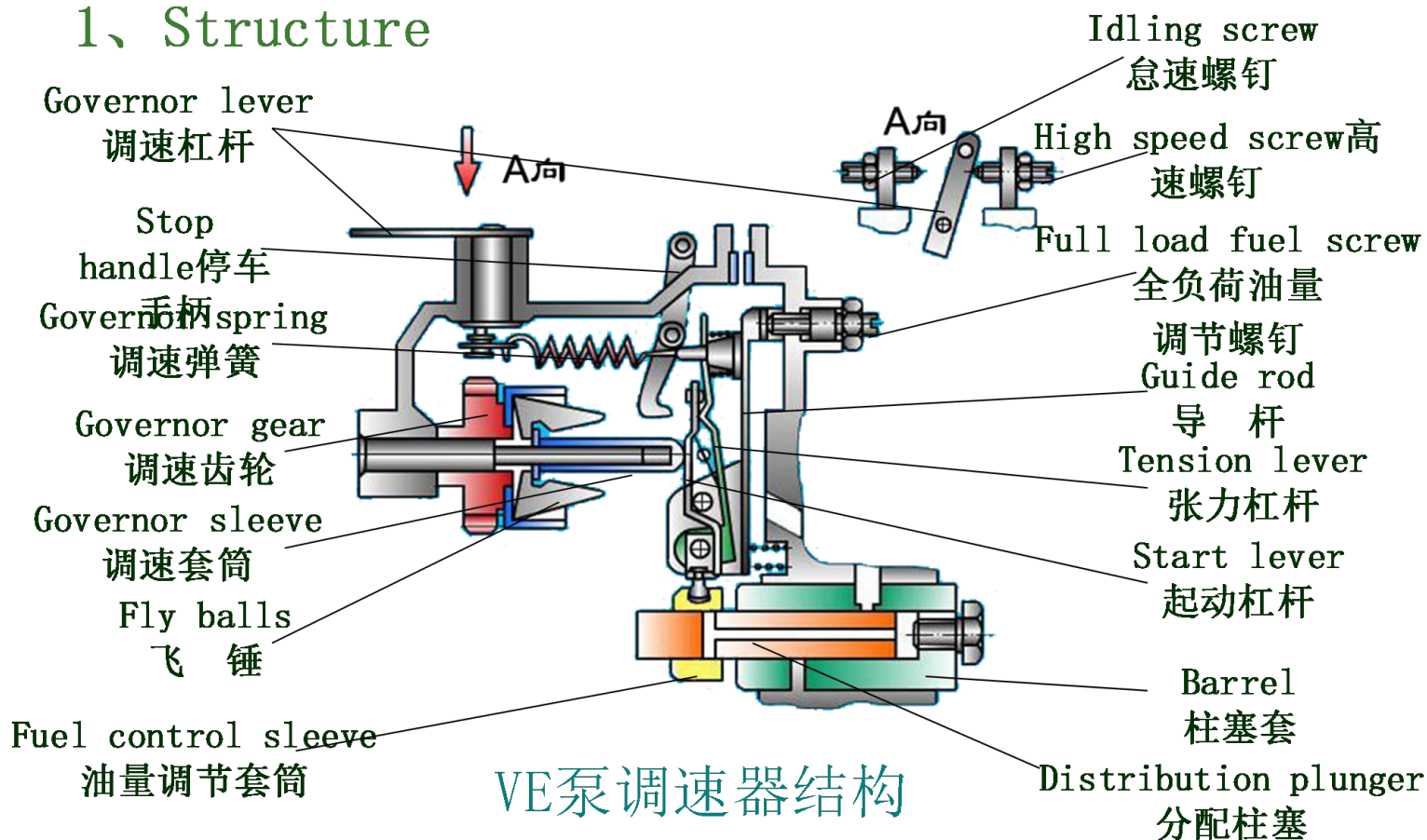
Flying balls are stuck by high speed spring seat and doesn't work

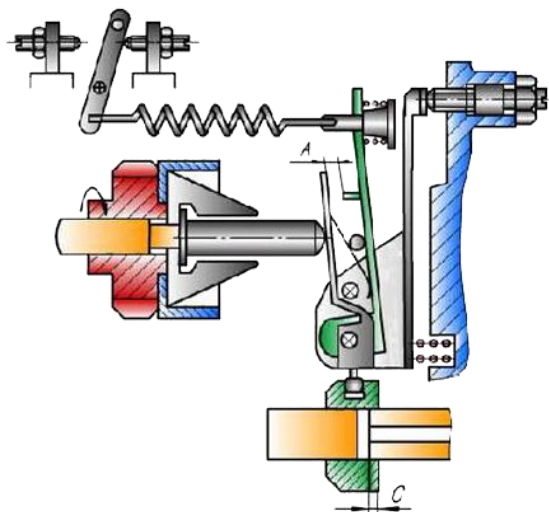
- **Engine speed above rated speed:**

Flying balls move further outwards and pull the fuel control rod back

5.6.2 Full-speed Governor (全速调速器)

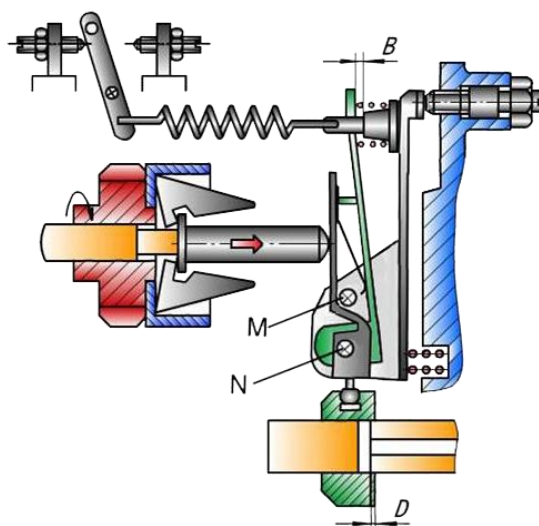
1、Structure





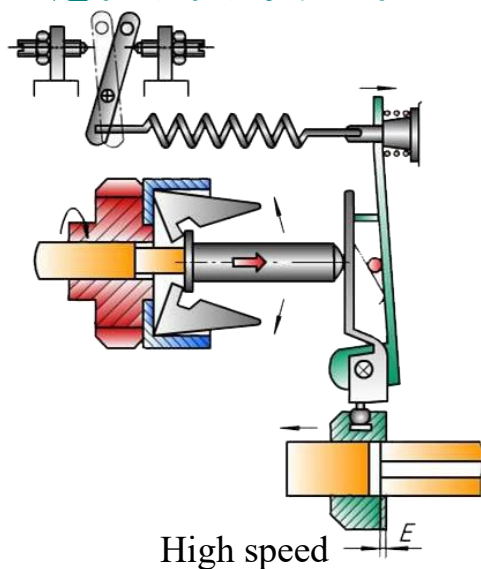
Engine start

VE泵调速器起动工况



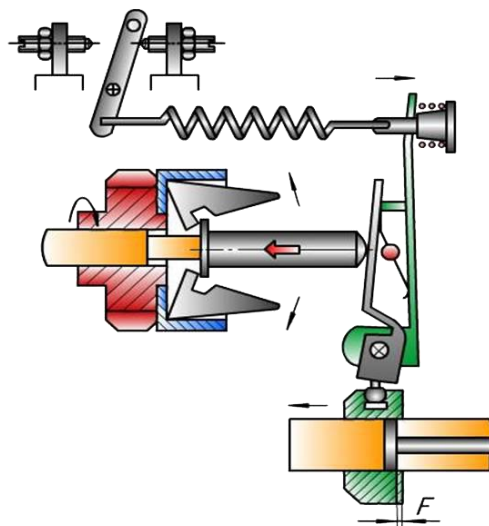
Idling

VE泵调速器怠速工况



High speed

VE泵调速器中高速工况



Overspeed

VE泵调速器超速工况

- **A**— Compression of start spring
- 启动弹簧压缩量
- **B**— Compression of idling spring
- 怠速弹簧压缩量
- **C**—Enriched fuel delivery position of engine start
- 启动加浓供油位置
- **D**—Fuel delivery position of idling
- 怠速工况供油位置
- **E**—Fuel delivery position of highest engine speed for part load
- 部分负荷最高转速供油位置
- **F**— Fuel delivery position of highest engine speed for full load
- 全负荷最高转速供油位置

5.6.3 Additional Features (附加装置)

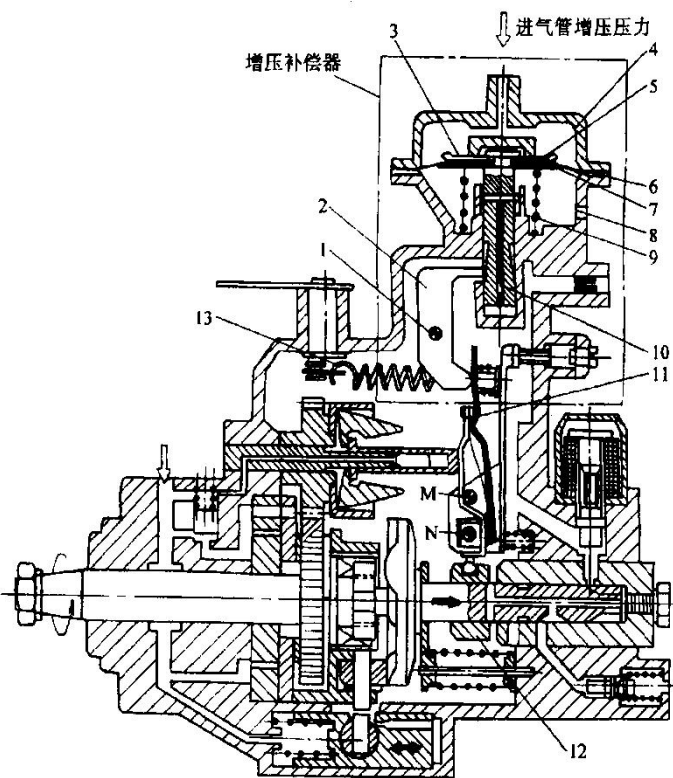


图 5-37 增压补偿器

1—销轴 2—补偿杠杆 3—膜片上支承板 4—补偿器盖 5—膜片 6—补偿器体 7—膜片下支承板
8—通气孔 9—弹簧 10—补偿器阀杆 11—张力杠杆 12—供油量调节套筒 13—调速弹簧

• Pressure Compensator

- ✓ Automatically regulate the amount of fuel delivery according to intake pressure when engine is boosted
- ✓ Enhance engine power, lower fuel consumption, lower smoke emission

增压补偿器:其作用是根据增压压力的大小,自动增减供油量,提高发动机功率,降低油耗,降低烟度。

Torque Calibration Device

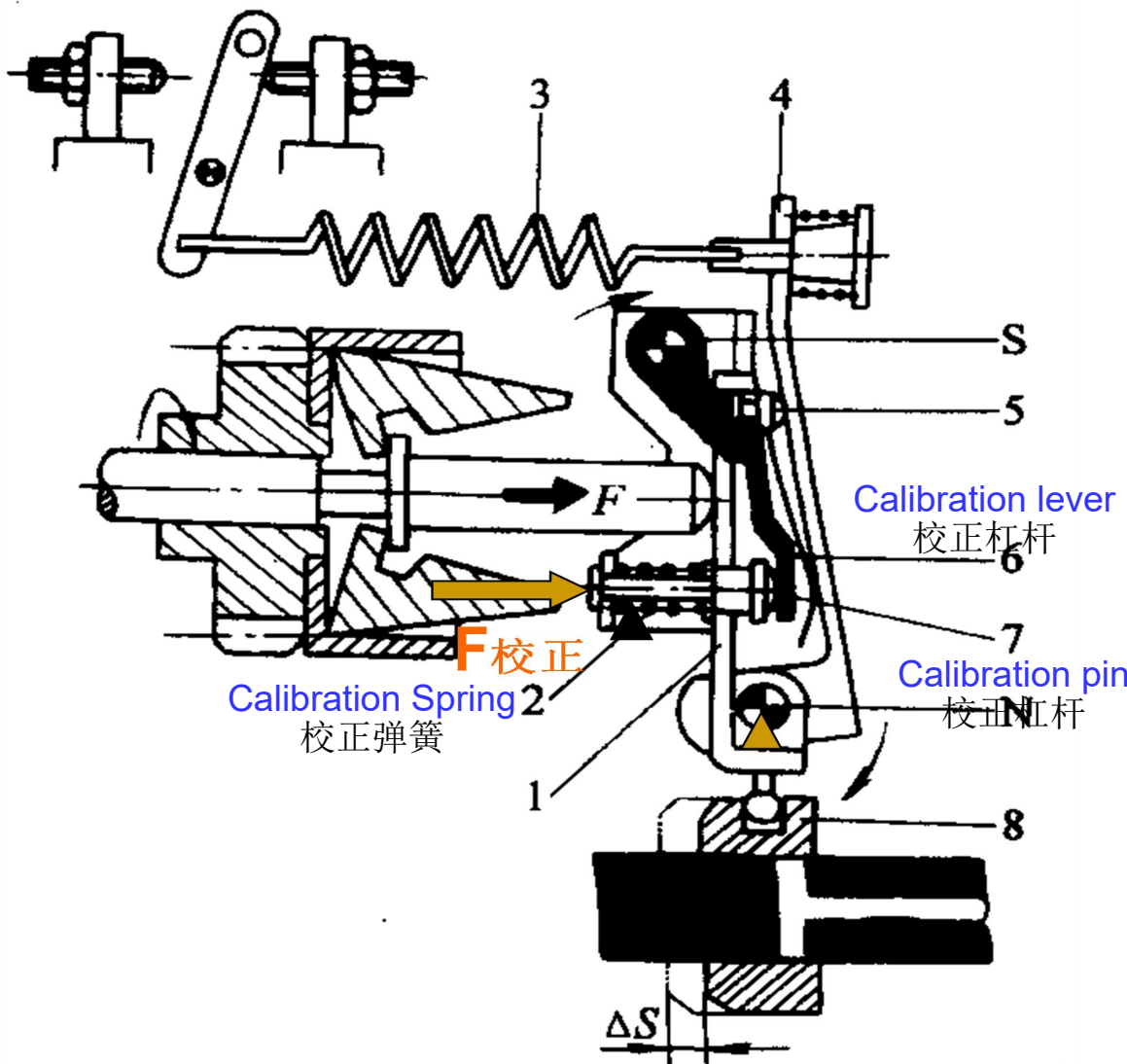
• Positive Calibration

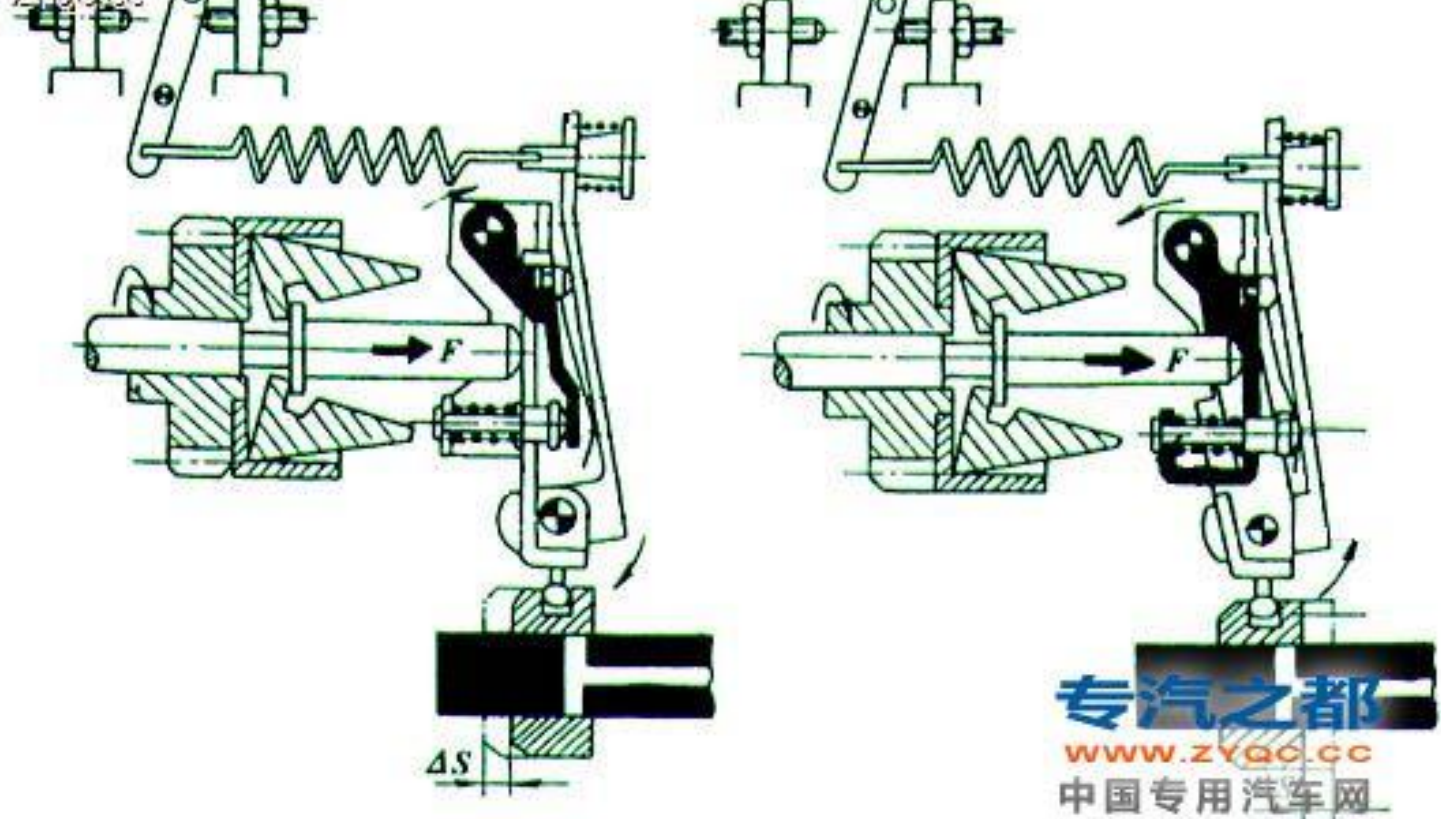
- ✓ Intake efficiency is best when engine runs at medium speed
- ✓ Increasing fuel delivery when engine runs at this speed could enhance engine torque output

发动机中间转速时气缸内的充气效率最高，可多供油使中间转速范围内输出转矩最高。这就意味着发动机从高速减速到中间转速时，喷油泵柱塞的有效压油行程在增大，供油量增加。

• Negative Calibration

- ✓ Decrease fuel delivery when engine transfers from high speed to low speed
- ✓ Eliminate smoke formation
- ✓ 负转矩校正可以防止柴油机低速时冒黑烟，即低速时齿杆行程减小，喷油泵的供油量减少。





转矩校正装置

根据需要可在VE型分配泵上装备正转矩校正或负转矩校正装置。正转矩校正可以改善柴油机高速范围内的转矩特性。当柴油机转速升高到校正转速时，随着转速继续升高，作用在起动杠杆上的飞锤离心力的轴向分力 F 对销轴 N 的力矩，逐渐超过校正弹簧的预紧力对校正杠杆的支点即挡销 5 的力矩，这时起动杠杆及销轴 S 开始绕销轴 N 向右摆动。与此同时，校正杠杆绕挡销顺时针方向转动，其下端通过校正销将校正弹簧压缩，直至校正销的大端靠在起动杠杆上为止，校正过程结束；负转矩校正可以防止柴油机低速时冒黑烟。在负转矩校正装置中，调速套筒的轴向分力 F 直接作用在转矩校正杠杆上，使校正杠杆靠在张力杠杆的挡销上，转矩校正销靠在张力杠杆的停驻点上。当柴油机转速升高时，调速套筒的轴向分力 F 增大。若轴向分力 F 对挡销的力矩大于校正弹簧的弹簧力对挡销的力矩，则使校正杠杆以挡销为支点逆时针方向转动，并通过销轴 S 带动起动杠杆绕销轴 N 向左摆动，球头销则向右拨动供油量调节套筒，增加供油量，从而实现柴油机在低速范围内随转速增加而自动增加供油量的负转矩校正。当校正杠杆靠在校正销大端上时，校正结束。

5.7 Electronic-controlled Injection System for Diesel Engine (柴油机电控燃油喷射系统)

● Function:

- ✓ Decrease diesel engine emissions 1、降低柴油机的排放;
- ✓ Enhance engine performance 2、改善柴油机的运转性能;
- ✓ Reduce fuel consumption 3、降低柴油机燃油消耗率。

● Advantages:

- ✓ Precise, sensitive, automatic adjustment with multi-function
- ✓ Less weight, smaller size, more compact
- ✓ Easy installation and connection, easier maintenance
- ✓ Self-diagnosis ability

- 1、精确、灵敏具有多功能的自动调节性能;
- 2、减轻重量, 缩小尺寸, 提高了柴油的紧凑性;
- 3、部件安装、连接方便, 提高了维修性;
- 4、具有自诊断、检测功能。

Classification

● Injection mass wise:

- ✓ Position controlled
- ✓ Time controlled

● Injection pressure wise:

- ✓ In-line pump
- ✓ Distributor pump
- ✓ Pump nozzle system
- ✓ Individual pump
- ✓ Common rail

5.7.1 Electronic Controlled Device (ECD) 电子控制装置

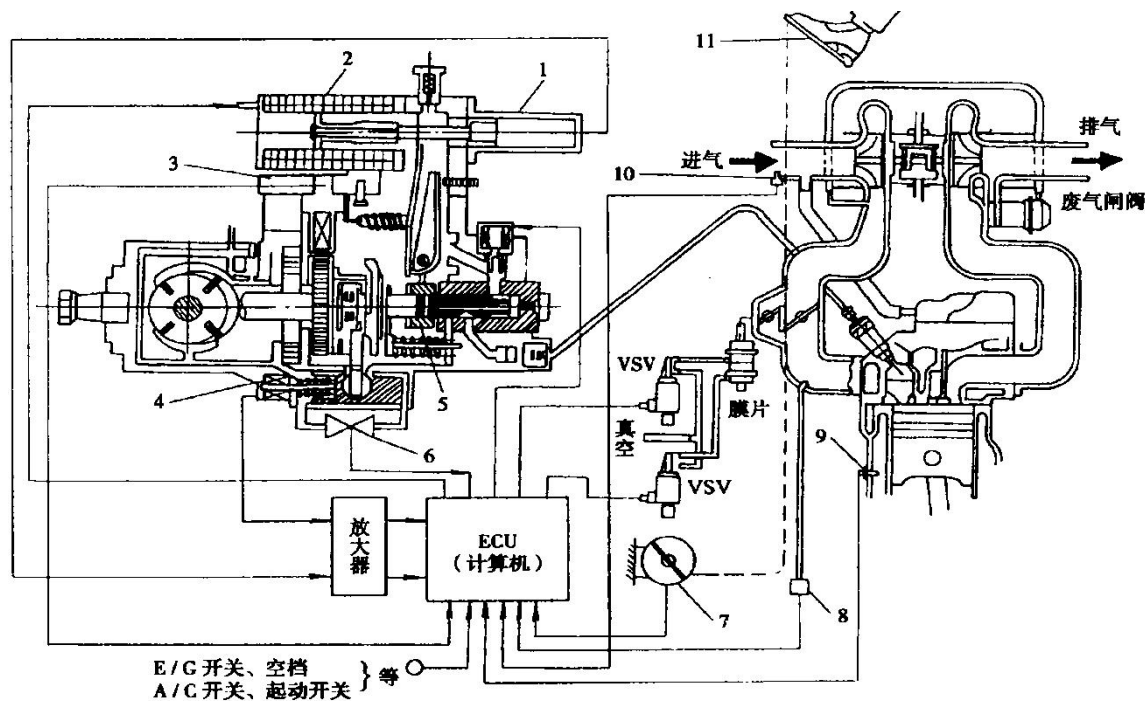


图 6-46 ECD 系统组成

1-供油量调节套筒位置传感器;2-供油量控制电磁阀;3-转速传感器;4-定时器位置传感器;5-供油量调节套筒;6-定时器控制阀;7-加速踏板位置传感器;8-进气压力传感器;9-冷却液温度传感器;10-进气温度传感器;11-加速踏板

- 1-Fuel control sleeve position sensor; 2-Solenoid valve for fuel delivery; 3-Rotational speed sensor; 4-Timer position sensor; 5-Fuel control sleeve ; 6-Timer control valve; 7-Position sensor for acceleration paddle; 8-intake pressure sensor; 9-coolant temperature sensor; 10-intake temperature sensor

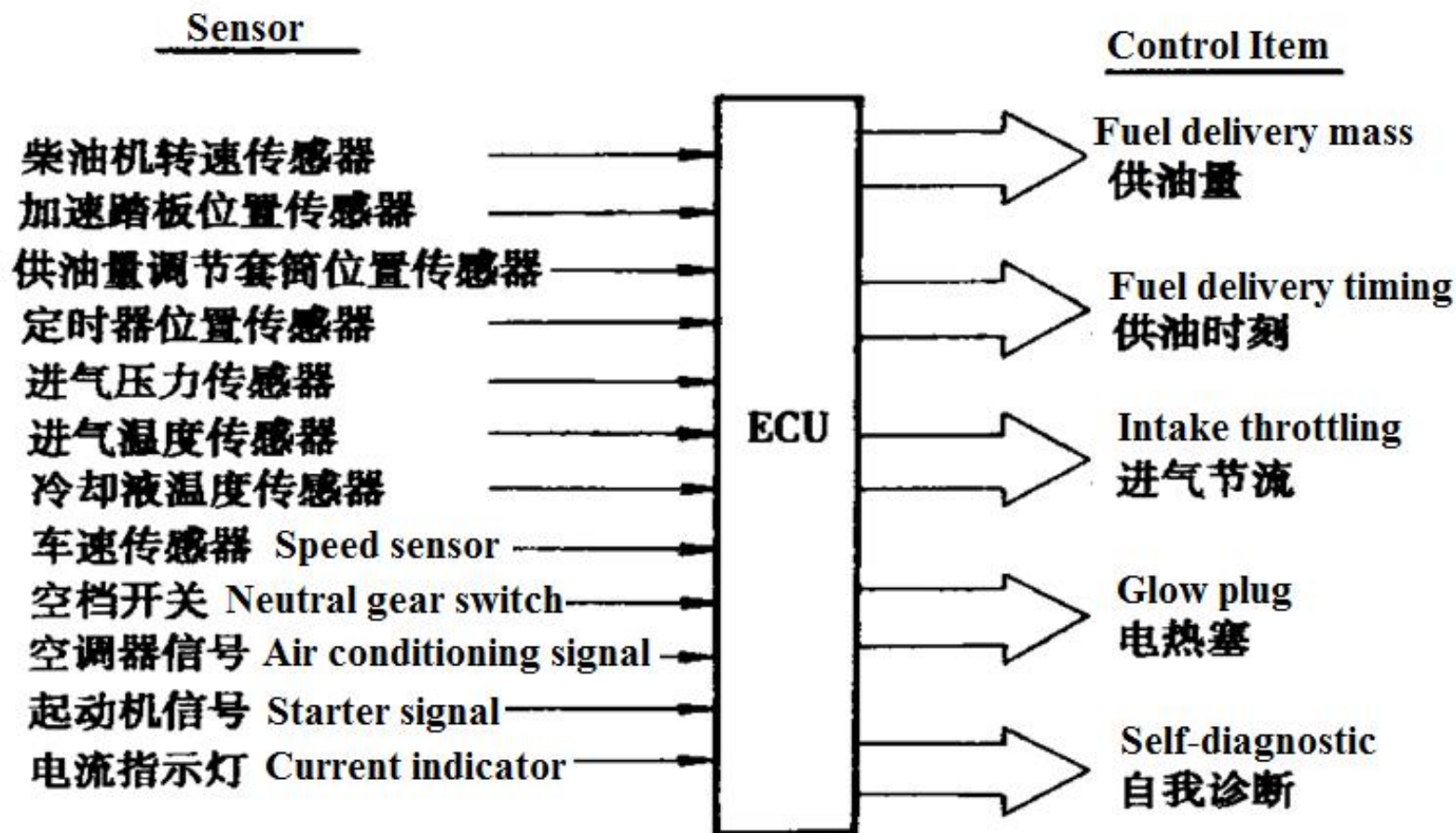
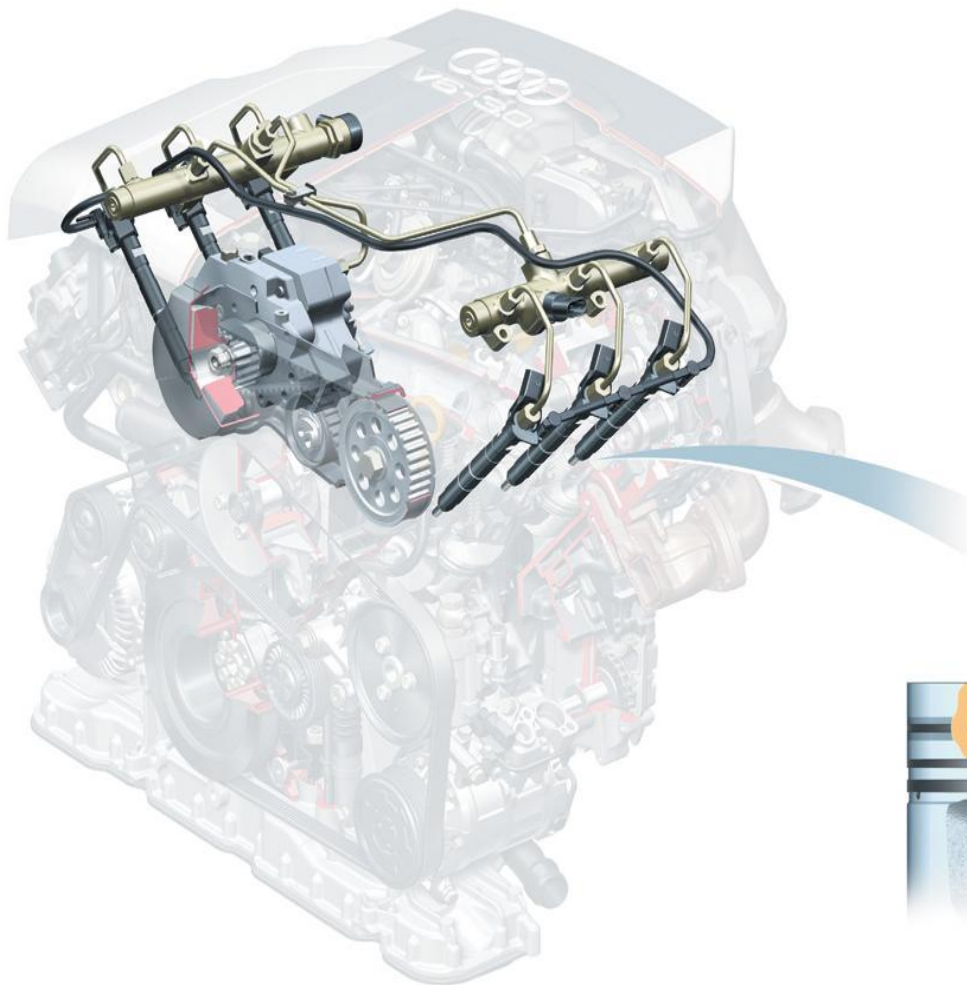
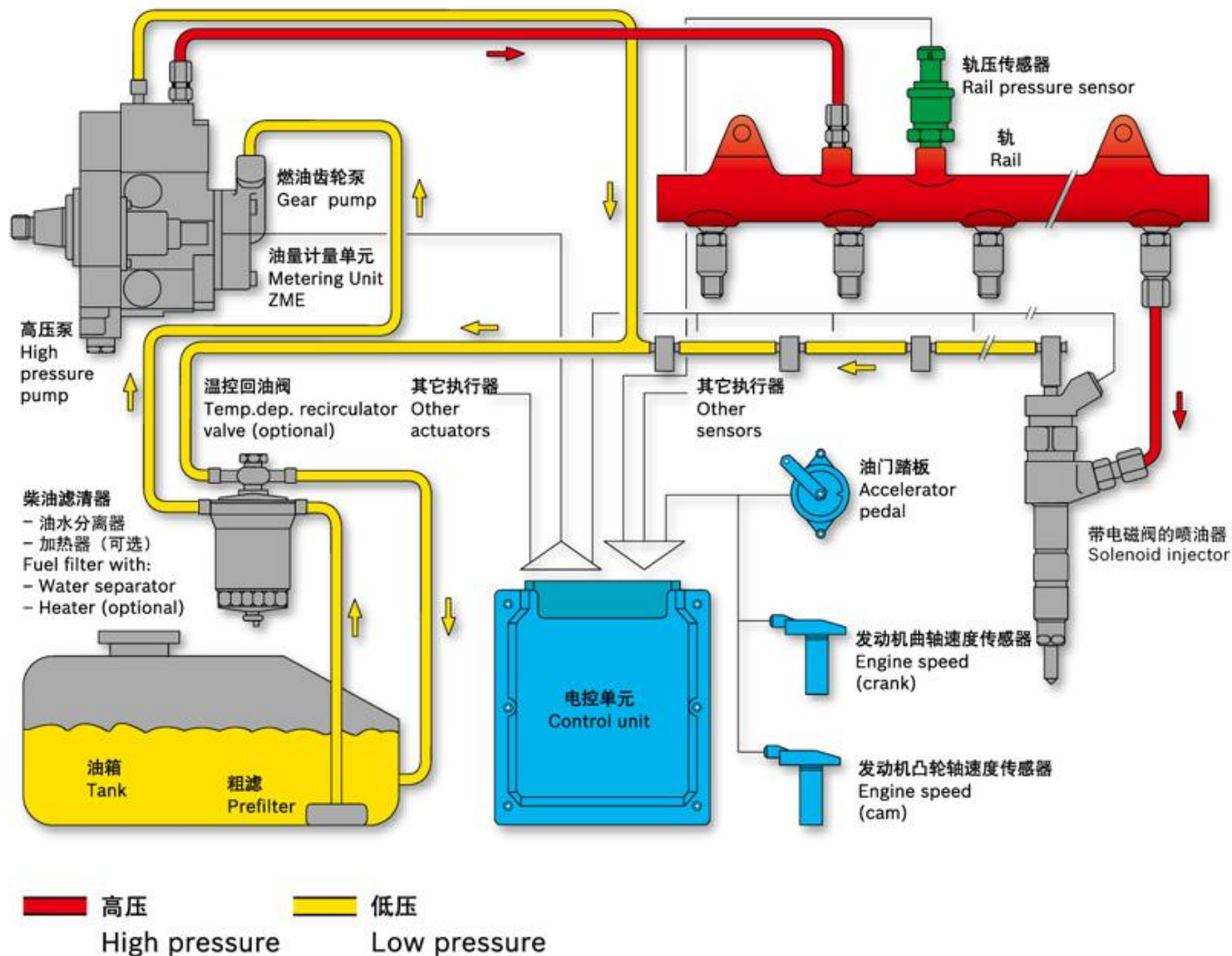


图 6-47 ECD 系统的控制项目及传感器

5.7.2 Common Rail Injection System (共轨系统)



- Freely selectable pressure within given limits
灵活的喷射压力
- Easy to optimize combustion
便于优化燃烧
- Possibility of multiple injections
可以实现多次喷射
- Reduce load on the pump drive
降低油泵负荷

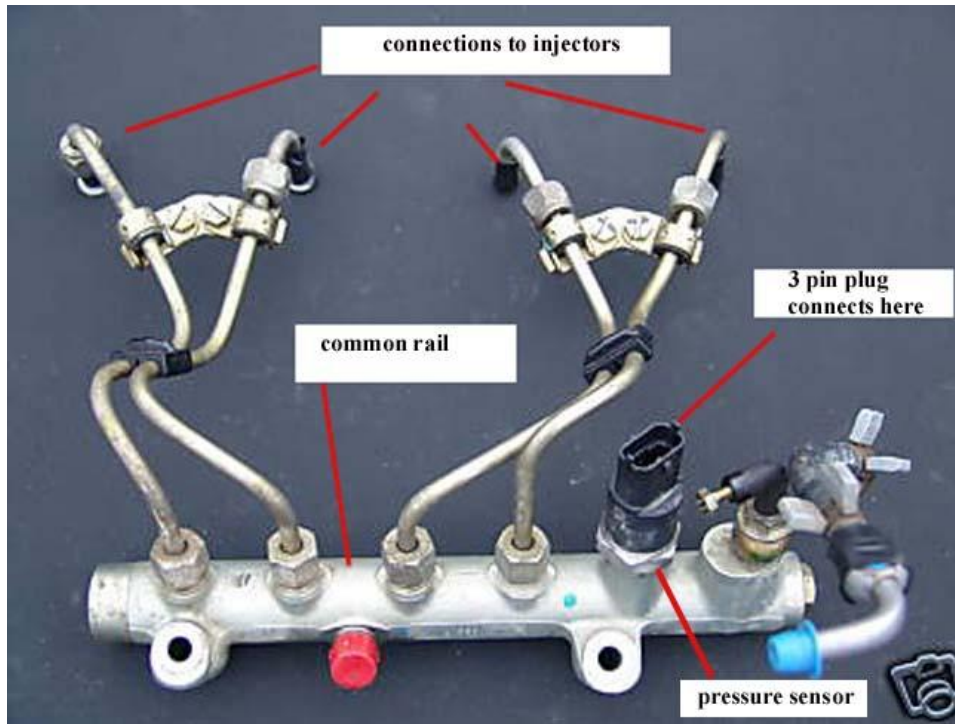


Control of Common Rail System



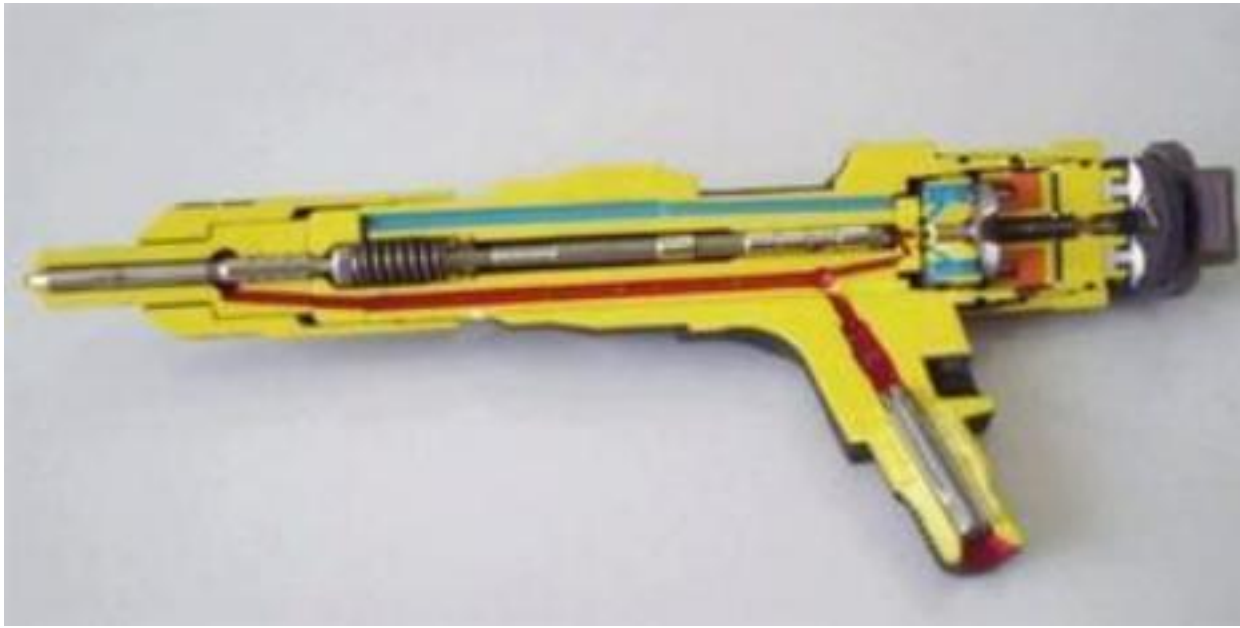
High pressure pump

- **High efficiency**
volumetric pump with energy use of 70% to 90% over pump speed range
- **High injection pressure**
- 200-1500 bar at maximum delivery



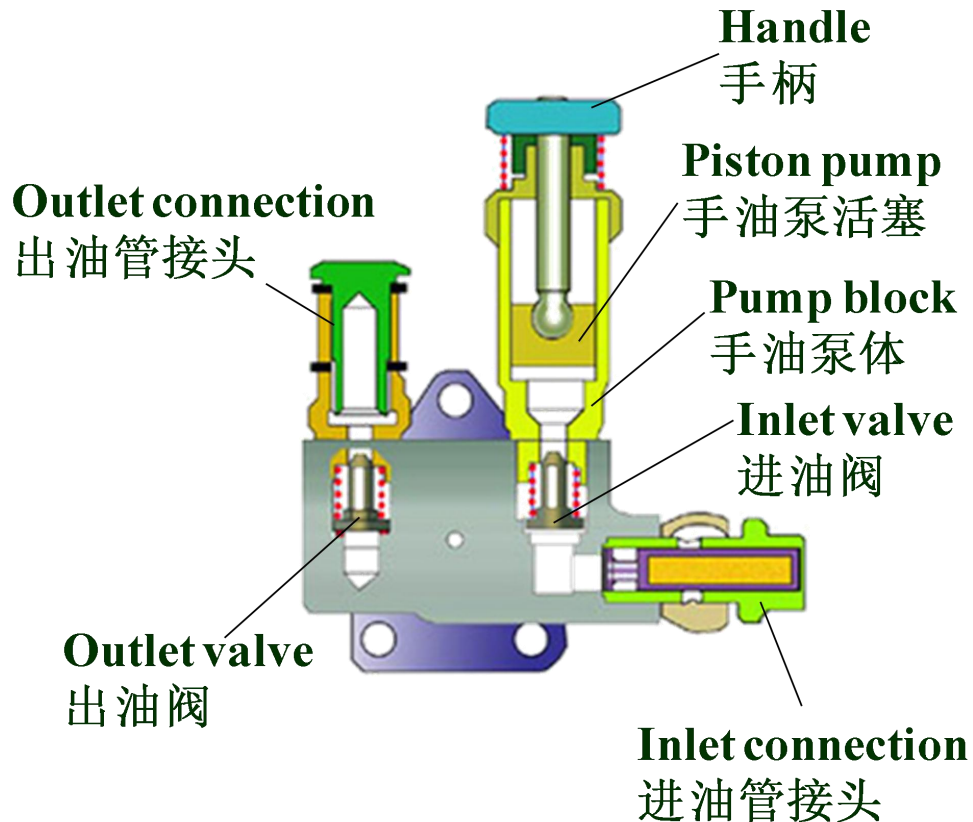
Rail

- Rail serves as a high-pressure reservoir
- Can be filled quickly to suppress pressure fluctuation during injection
- Rail is forged of welded(焊接) from drawn steel (冷拔钢)



- **Injector**
- Satisfy the automotive requirements regarding temperature and vibration
- Quick response, especially for multi-injection
- **Classification**
- Solenoid injector
 - ✓ higher electrical voltage
 - ✓ higher working temperature
- Piezoelectric injector(压电式)
 - ✓ Quick response
 - ✓ Lower working temperature

5.8 Auxiliary Devices (辅助装置)



• 5.8.1 Fuel Delivery Pump (输油泵)

• **Function:**

Guarantee normal diesel flow in low pressure channel, and deliver sufficient fuel to injection pump

The quantity of fuel delivered is 3~4 times as much as fuel consumption for full load

• **Classification:**

- ✓ Piston pump
- ✓ Rotor pump
- ✓ Sliding vane type
- ✓ Diaphragm type

Working Principle



Diaphragm type fuel pump
膜片式输油泵工作原理



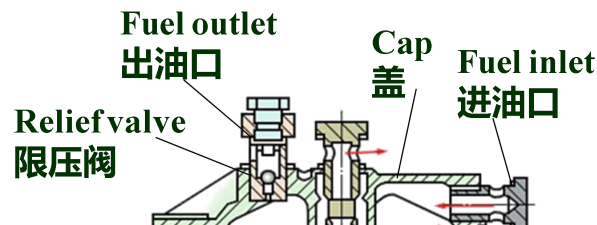
Piston fuel pump
活塞式输油泵工作原理

5.8.2 Diesel Filter (柴油滤清器)

Function:

- ✓ Remove the mechanical impurities and prevent the engine from wear and tear.
- ✓ Decrease the coupled pressure drop, low output, low efficiency.

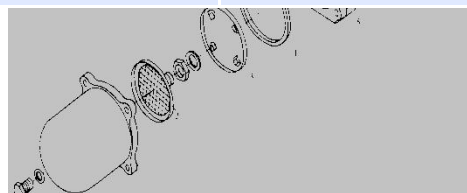
除去柴油机械杂质和油泥，防止柴油从柴油中析出，保护精密偶件的正常工作，降低油耗。



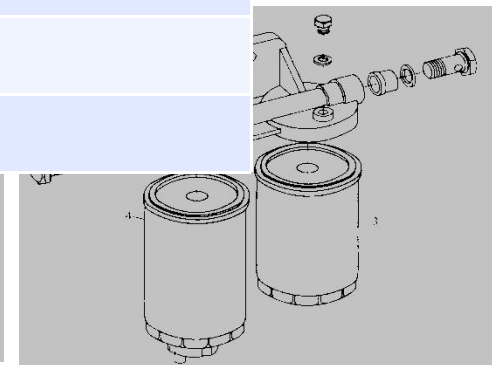
Video-5

element

rod



Diesel Filter (粗滤器)



Fine Filter (细滤器)

Personal homework

1. What is the Cetane Number?
2. What is the meaning of the commercial diesel number?
3. How to classify the combustion chamber of diesel engine?
4. How to classify the fuel injector?
5. How many kinds of diesel fuel injection pumps?
6. Why the in-line plunger pump is controlled by speed governor?
7. How many kinds of the electronic fuel injection system for diesel engine?
8. The consisting of common rail diesel fuel injection system.